Observer–Situation Constrained Multi-Agent Simulation Processes

Control under Fragmented Operational Truth

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Abstract

Regulated multi-agent logistics is commonly formulated as constrained control under partial observability: agents infer a latent operational state, optimize physical actions, and enforce safety, emission, or service constraints. This paper studies a complementary failure mode in which the operational problem is not merely hidden but *perspectivally fragmented*: incompatible reports about the same fact coexist because they originate from different observers, systems, timestamps, or operating situations. We introduce Observer–Situation Constrained Partially Observable Stochastic Games (OS-CPOSG) and the simulator-coupled subclass OS-CMASP. In these processes, the state contains a physical component and a formula-valued claim base indexed by observer–situation provenance; agents may take physical and epistemic actions; and critical physical actions are admissible only when supported by dominance-calibrated, consistency-checked certificates or routed through explicit exception acts.

The contribution is process-semantic rather than architectural. We prove containment results, an observer-collapse separation theorem showing that provenance erasure can induce linear regret under identical controller-visible context, a safety-budget separation corollary, and a certificate-soundness theorem for admissible critical actions. We then define Berth-1-Conflict, a minimal maritime benchmark and frozen-solver ablation protocol for testing whether provenance-preserving claim semantics separate from provenance-erased alternatives before full maritime-digital-twin scale-up. Emission and fairness translations are treated as secondary replay-certified validation claims. The resulting manuscript is a formal process-class proposal for control under fragmented operational truth, not a performance claim over a larger controller stack.

Keywords: multi-agent reinforcement learning; constrained stochastic games; simulator-coupled decision processes; epistemic action; truth maintenance; maritime logistics; fairness constraints; provenance-aware control.

Contents

Reader roadmap. Sections 1–7 contain the main paper: motivation, novelty boundary, related work, formal model, and core theorems. Sections 8–10 define the minimal maritime validation protocol. The appendices collect proof details, replay schemas, extended budget and fairness machinery, and the dominance-gated implementation witness.

Evidence status. The current paper is a theory-and-protocol manuscript. It intentionally contains no local performance claims. Numerical evidence enters only after the visible-context map, dominance order, replay schema, condition set, and paired-seed protocol are locked.

1 Introduction

Maritime coordination is usually written as a problem of constrained control under partial observability. A controller observes imperfect signals about berth status, vessel arrival, crane availability, weather, yard capacity, emissions exposure, and service delay, and then chooses physical actions subject to feasibility and budget constraints. This view connects cleanly to stochastic games [1, 2], decentralized partially observable control [3, 4], common-information control [5], constrained MDPs [6], and safe reinforcement learning [7–9]. It is also the natural language of many recent MARL systems for logistics and sustainability.

This paper studies a different failure mode. In port operations the difficulty is often not only that the state is hidden, but that the organization temporarily contains several incompatible descriptions of the same operational fact. An AIS forecast may predict on-time arrival while a pilot report contradicts it; a berth-management system may say that a berth is ready while a crane-maintenance feed says the berth is not operational; a regional weather source may clear a plan that a local sensor or human dispatcher rejects; or a strategic schedule may remain visible after a local operator has already overridden it. These are not merely independent noisy observations of a latent variable. They are provenance-bearing claims whose source, situation, freshness, and authority can determine whether an action is admissible.

A standard response is to fuse the reports into a posterior estimate. That response is appropriate when the only decision-relevant object is the latent physical state. It is not appropriate when the unresolved contradiction itself changes the feasible action set. For example, admitting a vessel under unresolved berth-readiness conflict may be unsafe; holding a vessel under a resolved but provenance-erased conflict may be unnecessarily costly; and escalating to a human dispatcher may be required precisely because a critical precondition lacks dominant support. In such cases the controller needs not only a belief about the world, but a representation of which claims support the action and which undefeated claims block it.

The research question is therefore:

> *Can constrained multi-agent control be formulated so that observer–situation provenance and unresolved contradictions are first-class state and feasibility objects, rather than metadata attached after a fused-state controller has already acted?*

We answer by defining Observer–Situation Constrained Partially Observable Stochastic Games (OS-CPOSG) and the simulator-coupled subclass Observer–Situation Constrained Multi-Agent Simulation Processes (OS-CMASP). The process state contains a physical component and a formula-valued claim base indexed by observer–situation provenance. Agents may take physical actions and epistemic actions such as query, verify, certify, defer, or escalate. Critical physical actions are admissible only when supported by a dominance-calibrated, consistency-checked certificate over the claim base, or when routed through an explicit exception act. Thus provenance is not an explanation attached to a policy; it is part of the process semantics that determines which actions are feasible.

This positioning matters because much nearby work already covers pieces of the space. Dec-POMDPs and POSGs already handle hidden state and local observations. Common-information methods and recent information-sharing MARL already study what can be learned when agents communicate or share histories. Epistemic planning already reasons about knowledge-changing actions. Truth-maintenance and belief-revision systems already maintain consistency under changing assumptions. Data-fusion and provenance systems already reason about conflicting sources. Safe RL and shielding already restrict unsafe actions. OS-CMASP is not presented as a replacement for those lines. Its narrower claim is that none of these, by itself, treats observer–situation-indexed contradiction components as simultaneous state variables, action-precondition objects, and audit certificates inside a constrained simulator-coupled stochastic game.

Contributions. The paper makes five contributions.

1. It defines OS-CPOSG and OS-CMASP, process classes in which formula-valued observer–situation claims, proposition-family dominance, contradiction components, epistemic repair actions, and certification-based admissibility are part of the formal model.
2. It states an anti-stacking boundary: the contribution is not a new high-level controller, belief module, LLM wrapper, or explanation interface. It is a change in process semantics; solver choice is intentionally secondary.
3. It proves an observer-collapse separation theorem: for a family of finite-horizon problems with identical controller-visible context and identical aggregated claim content, any policy invariant to observer–situation provenance suffers linear regret, while an observer-aware policy achieves zero regret. A safety-budget corollary shows that provenance erasure can also force a constraint tradeoff.
4. It proves certificate soundness for admissible critical actions: when an action is certified by a dominant, consistency-checked support set, the emitted explanation is a satisfiable witness for the action preconditions.
5. It specifies BERTH-1-CONFLICT, a minimal maritime benchmark and fixed-solver ablation protocol that tests whether provenance-preserving process semantics separate from provenance-erased alternatives before scaling to a full maritime digital twin.

Scope. The paper does not claim that every port-planning instance requires observer–situation semantics. If provenance never changes admissibility, certificates, exception requirements, or feasible value, then the active OS-CMASP regime is absent and a fused-state model may be sufficient. The validation protocol is therefore designed to identify where the semantics are operationally active, not to assert universal superiority.

2 Claim hierarchy and non-stacking scope

We position the paper as a process-semantics proposal rather than a new architecture proposal. To make that boundary explicit, table 1 separates the claims the paper is designed to support from secondary claims that require replay calibration.

Table 1: Claim hierarchy for the paper. Primary claims are semantic and solver-independent; metric claims are secondary.

Claim level	What is being claimed	What would weaken it
Primary semantic claim	Fragmented-truth state and certificate-based feasibility define a control problem that is not equivalent to a standard fused-state constrained POSG.	A provenance-erased fixed-solver controller matches the observer-aware controller on active-regime benchmarks without extra violations or interventions.
Primary theorem claim	Observer–situation provenance can be decision-relevant enough that erasing it induces linear regret or a zero-violation safety–service tradeoff.	The construction leaks hidden truth through visible state or relies on a dominance rule chosen after outcomes are known.
Secondary metric claim	Emissions and service-parity burdens can be lifted from the minimal benchmark into CH-MARL-style metrics through replay-certified envelopes.	Replay calibration is unstable, denominator distortion is large, or certified coefficients become non-positive under seed-shift stress.
Implementation claim	A dominance-gated wrapper can witness the semantics while keeping the solver frozen.	The observed gains disappear under solver-isolated ablations or are explained by controller changes rather than claim-state semantics.

Non-stacking criterion. A proposal would count as stacking, rather than as the contribution of this paper, if it leaves the process state as a single fused latent state and merely adds a belief module, explanation module, LLM interface, governance layer, or additional high-level planner. The present paper instead changes four objects: the state contains provenance-bearing claims, the action space includes epistemic repair, feasibility depends on certificates and exception acts, and the benchmark varies claim-state semantics while holding the solver fixed.

Definition 2.1 (Provenance-active regime)**.** An OS-CMASP instance is *provenance-active* for a policy class Π and scenario distribution $\mathcal{D}_{sc}$ if there exist two claim bases with identical aggregated formula content and identical controller-visible non-claim context, but different observer–situation provenance, such that at least one of the following changes: the admissible physical action set, the available dominant certificates for a critical action, the required exception act, or the feasible value for some policy in Π. If none of these changes, provenance may remain useful for audit but is not operationally active for that process instance and policy class.

This definition clarifies the active empirical regime. The maritime twin should be used to identify where provenance is operationally active, not to argue that every scenario requires the full observer–situation semantics.

3 Related work and precise delta

The main paper has to clear a high bar: many neighboring literatures already represent partial information, communication, knowledge, belief revision, source reliability, or safety constraints. The delta of OS-CMASP is therefore stated negatively and constructively. Negatively, the paper does not claim novelty for merely adding beliefs, communication, explanations, or safety filters to

CH-MARL. Constructively, it introduces a process semantics in which unresolved observer–situation contradictions directly affect state, admissibility, regret, and auditability.

3.1 POSGs, Dec-POMDPs, and information structure

Stochastic games and POSGs model sequential multi-agent decision-making under uncertainty [1, 2]. Dec-POMDPs formalize cooperative decentralized control under partial observability and are computationally hard even in finite-horizon settings [3, 4]. Common-information formulations show how shared histories can transform decentralized problems into coordinator-style belief problems [5]. Recent provable MARL work studies information sharing in POSGs and approximate common information as a route to tractability [10].

These models are directly relevant, but their ordinary object is a hidden state plus observations or shared histories. OS-CMASP can be viewed as changing the information structure itself: the state includes a claim base whose observer–situation labels cannot always be collapsed into a sufficient statistic without changing admissibility. The observer-collapse theorem is meant to isolate this point. It is not a claim that Dec-POMDPs cannot encode provenance in principle; any finite object can be appended to a state. The claim is that, once provenance-bearing contradiction is decision-relevant, it should be part of the process definition and benchmark contract rather than an informal feature added to a fused-state controller.

3.2 Constrained RL, safe RL, and shielding

Constrained MDPs provide the classical basis for cumulative budget constraints [6]. Safe RL and constrained policy optimization study how to optimize reward while limiting unsafe or budget-violating behavior [7, 8, 11]. Shielding methods synthesize monitors that prevent or correct unsafe actions according to formal specifications [9, 12]. EcoFair-CH-MARL uses related ideas in a maritime CH-MARL setting, with emissions budgets, fairness penalties, and a hierarchical controller [13, 14].

The distinction is that OS-CMASP does not add another safety layer above the policy. It changes what the shield or admissibility relation is allowed to inspect. A standard shield typically asks whether an action violates a state/property specification. In OS-CMASP, a critical action may be blocked because the support for its preconditions is defeated by a fresher, more authoritative, or situation-specific contrary claim. This makes the certificate and exception mechanism provenance-sensitive, not merely state-sensitive.

3.3 MARL solver families

Modern cooperative MARL includes centralized-training/decentralized-execution actor–critic methods [15, 16], value-decomposition methods [17, 18], and common-knowledge coordination policies [19]. These are solver families. They are important for later implementations, but they are not the novelty target. The benchmark protocol therefore freezes the solver and varies only claim-state semantics. If provenance-preserving and provenance-erased variants do not separate under the frozen solver and paired scenario bank, the process-semantics claim is not supported by that benchmark, regardless of whether a stronger architecture could be made to perform well.

3.4 Epistemic logic, epistemic planning, and interactive beliefs

Epistemic logic, dynamic epistemic logic, and common knowledge formalize how agents reason about knowledge and belief [20–23]. Epistemic planning uses knowledge-changing actions and event models to plan in multi-agent informational domains [24–27]. Interactive POMDPs model other agents inside sequential decision-making [28]. These lines are conceptually close because they treat information as part of agency rather than a post-hoc explanation.

The difference is operational coupling. OS-CMASP is not primarily about nested beliefs or epistemic goals. It is about regulated stochastic control where epistemic actions compete with physical actions under waiting costs, emissions proxies, safety constraints, and simulator-mediated dynamics. A query, certification, or escalation act matters because it changes which physical actions are admissible and which costs are incurred. The framework therefore combines epistemic support with constrained control rather than replacing control with epistemic planning.

3.5 Truth maintenance, belief revision, and non-monotonic support

Truth-maintenance systems track the reasons for beliefs and update belief sets when assumptions change [29–31]. AGM belief revision provides axioms for theory change under new information [32]. Abstract argumentation studies attack and acceptability relations among arguments [33]. These literatures are essential background for the contradiction and dominance components of the model.

OS-CMASP differs by making contradiction components control-relevant. Minimal contradiction decomposition, dominance, and consistency checks are not only used to maintain a coherent belief base. They enter the feasibility relation for critical physical actions and the regret analysis for provenance erasure. In that sense the claim base is not a passive knowledge store; it is part of the stochastic game's state and action-precondition semantics.

3.6 Provenance, trust, and multi-source data fusion

Database and workflow provenance record where data came from and how derived facts were produced [34–36]. Multi-source information fusion studies how to combine information from heterogeneous, possibly conflicting sources, including source reliability and conflict resolution [37–39]. Evidence-theoretic and subjective-logic approaches also provide tools for representing uncertainty and source trust [40–42].

The usual fusion objective is to infer or select a best estimate of the world. OS-CMASP instead preserves conflict when conflict is itself decision-relevant. A controller may need to hold, verify, or escalate not because the posterior probability is low, but because the dominant support certificate is absent or defeated. This is why the benchmark includes a provenance-erased condition: if fusion or aggregation is sufficient, erasure should not harm admissible performance in the paired setting.

3.7 Simulator-coupled processes and maritime logistics

R-CMASP is the closest process-level neighbor because it treats simulator coupling, structured roles, typed communication, and normative feasibility as part of a multi-agent decision process [43]. OS-CMASP extends that style of modeling in a different direction: the simulator-coupled process also carries a formula-valued claim base, dominance scheme, contradiction components, and certificate-based admissibility.

Maritime and logistics research provides the application substrate. Container terminal operations and empty-container repositioning have a long OR literature [44, 45]; MARL has been used for resource balancing in logistics networks [46]; and recent reinforcement-learning work studies empty-container repositioning [47]. These papers motivate the operational costs and constraints, but they do not make observer–situation provenance an action-feasibility object.

Table 2: Existing mechanisms already cover many pieces of the problem. The OS-CMASP delta is not any single ingredient; it is the process-level coupling of provenance-bearing claim state, contradiction-aware admissibility, epistemic exception actions, and fixed-solver semantic ablation.

Existing line	What it already handles	OS-CMASP delta
POSG / Dec-POMDP / common information	Hidden state, local observations, histories, information sharing, and shared-information reductions.	The claim base is part of state; observer–situation provenance is not silently fused away, and erasure is a formal ablation.
Constrained RL / shielding	Budgeted control, formal safety guards, and unsafe-action blocking.	Critical actions need dominant support certificates or explicit exception acts; the guard is provenance-sensitive.
Epistemic planning / I-POMDP	Knowledge-changing actions, nested beliefs, and models of other agents.	Epistemic actions are embedded in simulator-coupled constrained stochastic control with operational costs.
Truth maintenance / belief revision	Consistency maintenance, assumption tracking, and theory change.	Contradiction components affect physical-action admissibility, regret, and audit certificates.
Data fusion / provenance / trust	Source identity, reliability, evidence combination, and conflict resolution.	Conflict may be preserved because the unresolved conflict itself can require holding, verification, or escalation.
Maritime OR / logistics RL	Routing, capacity, repositioning, emissions proxies, and service metrics.	The maritime benchmark tests whether information conflict is an active process variable rather than ordinary noise.

Precise novelty statement. The novelty claim is not that observer labels, epistemic actions, provenance, or safety constraints are individually new. The claim is that their combination defines a distinct process semantics: a constrained simulator-coupled stochastic game whose state contains observer–situation-indexed formulas, whose critical actions require dominant consistency certificates or exception acts, and whose benchmark tests that semantics under a frozen solver.

4 Running example: berth-readiness under conflicting provenance

The main technical object is easier to see in a deliberately small maritime decision. A vessel approaches a berth, and the controller must decide whether to `admit` or `hold`. In a conventional partially observed formulation, the reports below would normally be converted into a single latent estimate such as $\mathbb{P}(ready(b) = 1 \mid h_t)$. That conversion is often sensible when reports are exchangeable noisy measurements. It is not sufficient when the operational rule asks a different question: *which report is allowed to certify a critical action under the current situation?*

Table 3: Illustrative claim base for a single berth-admission decision. The table separates measurement content from decision authority.

Observer	Situation	Formula	Operational role
Berth operating system	nominal slot	$ready(b)$	Supports admission under ordinary conditions, but may be defeated by fresher local evidence.
Crane maintenance feed	active outage	$\neg ready(b)$	Blocks admission when the outage is current and the crane family is safety-relevant.
AIS ETA service	approach window	$arrives\_soon(v)$	Affects holding cost and idling emissions, but does not certify berth readiness.
Harbor master	local confirmation	$ready(b)$	May dominate automated reports for berth-readiness certification when issued in the current situation.
Strategic plan	planning horizon	$ready(b)$	Useful for scheduling, but weaker than current local operational evidence.

The conflict is not only that $ready(b)$ and $\neg ready(b)$ coexist. Existing methods already have mechanisms for uncertainty, disagreement, and partial observation. The distinctive issue is that the contradiction is *typed by provenance*: the formulas are asserted by different observers, at different situation nodes, with different authority and freshness. A controller that collapses the table to a posterior probability may preserve enough information to choose a high expected-reward action, but it may lose the information required to decide whether that action is admissible, certifiable, or must be routed through an exception.

Example 4.1 (Two superficially identical aggregates)**.** Consider two claim bases with the same formulas and timestamps:

$$C^+ = \{(ready, \ell_H, 1, 0, \rho_1), (\neg ready, \ell_L, 1, 0, \rho_2)\},$$

$$C^- = \{(ready, \ell_L, 1, 0, \rho_1), (\neg ready, \ell_H, 1, 0, \rho_2)\},$$

where ℓ_H is dominant over ℓ_L for readiness claims. A provenance-erased controller sees the same aggregate in both cases. An observer–situation-aware controller sees that C^+ supports `admit` and C^- supports `hold`. The separation theorem in section 7 is the stochastic-game version of this example.

This example also clarifies what the paper is *not* claiming. It is not claiming that all multi-source conflicts require symbolic provenance, nor that every maritime decision should be passed through a theorem prover. The claim is narrower: in regulated, safety- or service-critical multi-agent control, there are regimes in which provenance-bearing contradictions are themselves part of the state of the control problem. In those regimes, observer–situation labels are not post-hoc explanation metadata; they affect admissibility, regret, and constraint tradeoffs.

Table 4: Why the berth example is not exhausted by common neighboring formalisms.

Neighboring line	What it captures	What remains exposed in the example
POMDP / POSG belief state	Hidden readiness and noisy observations [3, 48].	The posterior need not retain which observer–situation node can certify a critical action.
Common-information MARL	Shared histories and coordination-relevant information [5, 10, 19].	Shared information can still be a fused information state rather than a contradiction-bearing claim base.
Safe RL and shielding	Action suppression under safety constraints [7, 9, 12].	The shield condition is usually defined over states/specifications, not over undefeated provenance-indexed support.
Truth maintenance / belief revision	Inconsistent support and revision [29–32].	These are not normally embedded as feasibility objects inside constrained stochastic games.
Data provenance and fusion	Lineage, source conflict, and fusion policies [34, 35, 39].	Fusion is usually upstream of control, while here unresolved provenance can directly affect action admissibility.

The running example will be instantiated as Berth-1-Conflict in section 8. That benchmark is intentionally small: it is not a port-operations benchmark, but a test of the paper's process-semantic claim under a frozen solver.

5 Notation and modeling commitments

OS-CMASP separates objects that are frequently collapsed in applied MARL pipelines: the physical simulator state, the controller-visible physical context, the claim base, the dominance scheme, and the admissibility relation. This separation is the main novelty boundary. If these objects are collapsed before decision, the paper reduces to a familiar partially observed constrained-control problem.

Table 5: Core notation and modeling role.

Symbol	Meaning
X, x_t	Latent physical state and its value at time t. In maritime replay this may include berth status, queue, vessel, weather, crane, and disruption variables.
$\chi : X \to \mathcal{Y}$	Controller-visible context map. It includes ordinary non-claim context available to all compared policies, and excludes the hidden truth variable used in the separation theorem.
$\mathcal{O}$, Σ, L	Observer set, situation set, and observer–situation carrier $L = \mathcal{O} \times \Sigma$.
$\mathcal{L}_\Phi$, Cons	Formula language and sound consistency predicate. The minimal benchmark uses literals; richer maritime assertions may use SAT/SMT-backed formulas.
C_t	Formula-valued claim base containing formulas, observer–situation labels, credibility, timestamps, and provenance.
$\mathcal{D}$	Proposition-family dominance scheme. It encodes pre-registered source authority, freshness, context priority, and tie rules.
A_i^p, A_i^e	Physical and epistemic action spaces for agent i. Epistemic actions include verification, deferral, escalation, and certification.
$\mathcal{F}$	Admissibility relation coupling physical feasibility, dominant support certificates, contradiction mass, and exception acts.
$\mathrm{Agg}(C)$	Aggregation that erases observer–situation labels and provenance; used to define provenance-erased baselines.

Table 6: Semantic roles separated by the process. The final column states the failure mode if the role is collapsed.

Role	Object	Failure mode when collapsed
Physical dynamics	x_t and P	The simulator may predict feasible trajectories, but cannot by itself determine which report is admissible support.
Visible context	$\chi(x_t)$	Hidden truth may leak into the baseline comparison, invalidating the observer-collapse theorem.
Operational claims	C_t	Contradictions are prematurely fused into a single estimate, losing provenance needed for certification.
Defeat / priority	$\mathcal{D}$	Authority and freshness are tuned informally after outcomes, converting the benchmark into an unregistered heuristic.
Admissibility	$\mathcal{F}$	Safety and auditability become external wrappers rather than state-dependent feasibility constraints.

The following commitments are fixed before any experiment.

1. **Visible-context commitment.** The map χ is defined before running the ablations and is used identically by all non-oracle policies. In Berth-1-Conflict, χ may include vessel class, queue state, weather regime, disruption family, operating mode, time bucket, berth slot, and ETA bin, but not latent readiness, crane safety, weather admissibility, or claim provenance.

2. **Dominance commitment.** The dominance scheme $\mathcal{D}$ is registered by proposition family. It cannot be tuned after seeing paired deltas.

3. **Solver-isolation commitment.** The policy code, action costs, episode bank, and random seeds are fixed across all ablations. Conditions change only the claim-state semantics.

4. **Evidence-status commitment.** Synthetic smoke tests are pipeline checks, not evidence. Paper evidence begins only when the same frozen interface is run on a replay bank or a declared synthetic benchmark with locked parameters.

These commitments make the model stricter than an architecture proposal. A new controller can be plugged in later, but the paper's first claim must survive when the solver is held fixed.

6 Observer–Situation Constrained Process Model

6.1 Observer–situation carrier, formula language, and claims

Definition 6.1 (Observer–situation carrier)**.** Let $(\mathcal{O}, \preceq_{\mathcal{O}})$ be a finite complete lattice of observers and $(\Sigma, \preceq_\Sigma)$ a finite complete lattice of situations. Define

$$L = \mathcal{O} \times \Sigma, \qquad (o, \sigma) \preceq (o', \sigma') \iff o \preceq_{\mathcal{O}} o' \text{ and } \sigma \preceq_\Sigma \sigma'.$$

Then L is a finite complete lattice under the product order.

The observer order need not be a single global hierarchy. A harbor master may dominate for berth admission; a maintenance feed may dominate for crane availability; an approved weather source may dominate for safety windows; and a human dispatcher may dominate when an override is explicitly recorded. For this reason, dominance is defined by proposition family below rather than by a single universal source order.

Definition 6.2 (Formula language and consistency oracle)**.** Let Φ be a finite propositional vocabulary and let $\mathcal{L}_\Phi$ be a finite set of propositional formulas over Φ closed under negation for the formulas used as action preconditions. Let

$$\mathsf{Cons} : 2^{\mathcal{L}_\Phi} \to \{0, 1\}$$

be a sound consistency predicate, where $\mathsf{Cons}(B) = 1$ means that the finite formula set B is propositionally satisfiable. We write $\mathsf{Incon}(B) = 1 - \mathsf{Cons}(B)$. For the literal-only benchmark, $\mathcal{L}_\Phi = \mathrm{Lit}(\Phi)$ and Cons is the no-complement-pair test. For richer operational schemas, Cons can be implemented by a SAT or SMT solver.

Definition 6.3 (Claim base)**.** A claim record is

$$c = (\varphi, \ell, w, \tau, \rho) \in \mathcal{L}_\Phi \times L \times [0, 1] \times \mathbb{T} \times \mathcal{P},$$

where φ is the asserted formula, ℓ is the observer–situation location, w is credibility, τ is a timestamp, and ρ is provenance metadata. A finite claim base C is a finite set of such records. For a decision node $\lambda \in L$, the supported theory is

$$\mathrm{Th}_\lambda(C) = \{\varphi : \exists(\varphi, \ell, w, \tau, \rho) \in C \text{ with } \ell \preceq \lambda\}.$$

The claim base is not a belief state in the standard filtering sense. It is allowed to contain incompatible formulas. Its purpose is to preserve the structure of incompatibility until a dominance rule, an epistemic action, or an exception path resolves its decision relevance.

Definition 6.4 (Proposition-family dominance scheme)**.** Let $\mathcal{Q}$ be a finite set of proposition families and let

$$\mathsf{fam} : \mathcal{L}_\Phi \to \mathcal{Q}$$

map each formula to the operational fact-family it concerns, such as berth readiness, ETA, crane availability, weather admissibility, customs release, or emissions feasibility. A dominance scheme is a family of preorders

$$\mathcal{D} = \{\trianglerighteq_q : q \in \mathcal{Q}\}$$

over claim records with family q. The strict part is denoted $\triangleright_q$. The preorder may encode source authority, freshness, calibrated reliability, regulatory priority, or situation specificity. It is a process parameter and must be fixed before evaluation. For a finite set of claims B of family q, define its undefeated elements as

$$\mathrm{Max}_{\trianglerighteq_q}(B) = \{c \in B : \nexists c' \in B \text{ such that } c' \triangleright_q c\}.$$

Definition 6.5 (Contradiction graph)**.** Given C, define $G_\perp(C) = (V, E)$ with $V = C$. For distinct claims $c = (\varphi, \ell, w, \tau, \rho)$ and $c' = (\psi, \ell', w', \tau', \rho')$, include edge $\{c, c'\} \in E$ iff

$$\ell \text{ and } \ell' \text{ are comparable in } L \quad \text{and} \quad \mathsf{Incon}(\{\varphi, \psi\}) = 1.$$

Let $\mathrm{MCC}(C)$ denote the non-singleton connected components of $G_\perp(C)$.

Remark 6.6 (Relation to truth maintenance)*.* The contradiction graph resembles a truth-maintenance dependency structure [29–31], but its role is different. Here the contradiction components are part of the stochastic-game state and can enter rewards, constraints, admissibility, and audit traces. The formal object is therefore not a standalone belief-revision system; it is a constrained control process with embedded claim maintenance.

6.2 The OS-CPOSG

Definition 6.7 (Observer–Situation Constrained POSG)**.** An Observer–Situation Constrained Partially Observable Stochastic Game (OS-CPOSG) is a tuple

$$\mathcal{G} = \langle N, X, L, \mathcal{L}_\Phi, \mathcal{D}, \{A_i^p, A_i^e, \Omega_i\}_{i \in N}, P, Z, \Gamma, r, \mathbf{g}, \mathcal{F}, \gamma \rangle,$$

where N is the agent set; X is the physical state space; $\mathcal{D}$ is a proposition-family dominance scheme; A_i^p and A_i^e are physical and epistemic action sets; Ω_i is the observation set; P is the physical transition kernel; Z is the observation kernel; Γ maps observations and epistemic acts to new claim records; r is the operational reward; $\mathbf{g} = (g_1, \ldots, g_m)$ are cumulative constraint costs; $\mathcal{F}$ is an admissibility relation; and $\gamma \in (0, 1)$ is a discount factor.

The augmented state is

$$s_t = (x_t, C_t) \in X \times 2^{\mathcal{L}_\Phi \times L \times [0,1] \times \mathbb{T} \times \mathcal{P}}.$$

A joint action splits into physical and epistemic components,

$$a_t = (a_t^p, a_t^e), \qquad a_t^p \in \prod_i A_i^p, \quad a_t^e \in \prod_i A_i^e.$$

Physical dynamics and observations evolve as

$$x_{t+1} \sim P(\cdot \mid x_t, a_t), \qquad o_{t+1} \sim Z(\cdot \mid x_{t+1}, a_t),$$

and the claim base updates by

$$C_{t+1} = \mathrm{RBP}\left(C_t \cup \Gamma(o_{t+1}, a_t^e)\right), \qquad \mathcal{K}_{t+1} = \mathrm{MCC}(C_{t+1}).$$

The epistemic action space is deliberately part of the game rather than an off-line preprocessing step. A `verify` action can request a new claim from an authoritative observer; `defer` can postpone a physical action while preserving the unresolved contradiction; `escalate` can route the conflict to a human or institutional authority; and `certify` can emit a support witness for an action. Different applications may instantiate these actions differently, but the process class treats them as actions with costs and transition effects.

Definition 6.8 (Simulator-coupled subclass)**.** An Observer–Situation Constrained Multi-Agent Simulation Process (OS-CMASP) is an OS-CPOSG whose physical transition kernel is mediated by a simulator family $\mathcal{S}$:

$$x_{t+1} \sim P_{\mathcal{S}}(\cdot \mid x_t, a_t, \xi_t),$$

where ξ_t denotes exogenous simulator input such as traffic demand, weather, disruption state, scenario seed, or simulator-version state.

The simulator-coupled definition is the maritime target. The simulator can remain a black-box digital twin, but the claim state is not black-box: it is an explicit part of the process state and of the ablation contract.

6.3 Dominant support, certification, and admissibility

For each agent i and physical action a_i^p, let $\mathrm{Pre}_i(a_i^p) \subseteq \mathcal{L}_\Phi$ be the action precondition set. Let $\lambda_i : X \to L$ map physical states to the observer–situation decision node at which agent i evaluates support.

Definition 6.9 (Dominant relevant claim set)**.** For action a_i^p at state (x, C), let

$$Q_i(a_i^p) = \{\mathsf{fam}(\varphi) : \varphi \in \mathrm{Pre}_i(a_i^p)\}$$

be the relevant proposition families. For each $q \in Q_i(a_i^p)$, define

$$B_{i,q}(x, C) = \{c = (\varphi, \ell, w, \tau, \rho) \in C : \ell \preceq \lambda_i(x),\ \mathsf{fam}(\varphi) = q\}.$$

The dominant relevant claim set is

$$\mathsf{DRel}_i(x, C, a_i^p) = \bigcup_{q \in Q_i(a_i^p)} \mathrm{Max}_{\trianglerighteq_q}(B_{i,q}(x, C)).$$

Definition 6.10 (Dominant support certificate)**.** A dominant support certificate for action a_i^p at (x, C) is the set $S = \mathsf{DRel}_i(x, C, a_i^p)$, provided that

$$\mathsf{Cons}(\mathrm{Th}_{\lambda_i(x)}(S)) = 1$$

and

$$\mathrm{Th}_{\lambda_i(x)}(S) \models \varphi \qquad \text{for every } \varphi \in \mathrm{Pre}_i(a_i^p).$$

The family of such certificates is denoted $\mathsf{DCert}_i(x, C, a_i^p)$. Since DRel_i is determined by (x, C, a_i^p) and $\mathcal{D}$, DCert_i is either the singleton $\{\mathsf{DRel}_i(x, C, a_i^p)\}$ or empty.

Remark 6.11 (No cherry-picking). The dominant certificate definition prevents an agent from certifying a critical action by selecting a convenient satisfiable subset while ignoring a more dominant contradictory record. If the undefeated relevant claims are mutually inconsistent, certification fails and the action must be routed through an epistemic exception act or made inadmissible. This is the point at which the process differs from merely attaching explanations to a conventional MARL policy.

Definition 6.12 (Contradiction mass)**.** Let $\mathsf{Rel}_i(x, C, a_i^p)$ be the subset of claims in C whose proposition family is in $Q_i(a_i^p)$. Given nonnegative component weights $\omega(K)$,

$$\kappa_i(x, C, a_i^p) = \sum_{K \in \mathrm{MCC}(C)} \omega(K)\, \mathbf{1}[K \cap \mathsf{Rel}_i(x, C, a_i^p) \neq \varnothing].$$

Definition 6.13 (Contradiction-aware admissibility)**.** Let $A_i^{\mathrm{crit}} \subseteq A_i^p$ be the critical physical actions and let $\mathsf{Exc}_i(a_i^p) \subseteq A_i^e$ be exception acts, such as `verify`, `defer`, or `escalate`. A joint action $a = (a^p, a^e)$ is admissible at (x, C), written $\mathcal{F}(x, C, a) = 1$, iff physical feasibility holds and, for every i with $a_i^p \in A_i^{\mathrm{crit}}$, either

$$\mathsf{DCert}_i(x, C, a_i^p) \neq \varnothing \quad \text{and} \quad \kappa_i(x, C, a_i^p) \leq \eta_i,$$

or $a_i^e \in \mathsf{Exc}_i(a_i^p)$.

The optimization problem is

$$\max_{\pi}\ \mathbb{E}_\pi \sum_{t=0}^{T-1} \gamma^t \left[r(x_t, a_t^p) - \lambda_\perp \sum_i \kappa_i(x_t, C_t, a_{i,t}^p) - \lambda_e \sum_i c_e(a_{i,t}^e) \right]$$

subject to cumulative constraints

$$\mathbb{E}_\pi \sum_{t=0}^{T-1} g_j(x_t, a_t^p) \leq B_j, \qquad j = 1, \ldots, m,$$

and admissibility $\mathcal{F}(x_t, C_t, a_t) = 1$ for all t.

Remark 6.14 (Why this is not a stack). A stacked controller would keep the underlying process fixed and add a provenance layer, a shield, or an explanation module around it. OS-CMASP changes the process itself: claim records are state variables, epistemic acts are actions, and certification is part of feasibility. A solver can be simple or complex, but the distinction being tested is whether the process semantics are preserved or erased.

7 Theoretical properties

7.1 Containment and boundary cases

Proposition 7.1 (Containment)**.** *The proposed model contains or projects to familiar process classes under parameter restrictions.*

1. *If L is a singleton, $A_i^e = \{\varnothing\}$, $\mathcal{D}$ is vacuous, and $\mathcal{F}$ is independent of C_t, then* OS-CPOSG *reduces to a constrained POSG; in the cooperative case it reduces to a constrained Dec-POMDP.*

2. *If the physical kernel is simulator-mediated and admissibility contains only exogenous regulatory or organizational rules, then* OS-CMASP *projects to a simulator-coupled norm-governed process.*

3. *If policies are restricted to the two-timescale hierarchical controller family used by CH-MARL/EcoFair, then the result is a controller family over an* OS-CMASP *environment rather than a new process class.*

Proof. The restrictions remove observer–situation structure, dominance, epistemic actions, and claim-dependent feasibility. What remains is a standard constrained stochastic game or its cooperative partial-observation specialization. Simulator coupling is retained by choosing $P = P_{\mathcal{S}}$. Restricting the policy family affects the solver class but not the environment semantics. $\square$

The containment result is useful for two reasons. First, it prevents the formalism from becoming an unrelated notation system: it reduces to standard objects when the proposed semantics are inactive. Second, it makes ablation interpretation precise: provenance-erased and labels-without-gating conditions are not weaker neural architectures, but projections of the process state and feasibility relation.

7.2 Observer-collapse separation

The central theorem must avoid a common loophole: if the physical state differs across cases and the policy can observe that difference, hidden truth may leak through ordinary non-claim features. We therefore make the controller-visible physical context explicit.

Definition 7.2 (Visible context and provenance-erased policies)**.** Let $\chi : X \to \mathcal{Y}$ be the controller-visible physical context map. Let $\mathrm{Agg}(C)$ be the multiset obtained from C by discarding observer–situation locations and provenance while retaining asserted formulas, credibilities, and timestamps. A policy π is provenance-erased if

$$\chi(x) = \chi(x') \text{ and } \mathrm{Agg}(C) = \mathrm{Agg}(C') \quad \Rightarrow \quad \pi(\cdot \mid x, C) = \pi(\cdot \mid x', C').$$

The class of such policies is denoted Π_{erase}.

Theorem 7.3 (Observer-collapse separation)**.** *For every horizon $T \geq 1$, there exists a finite-horizon* OS-CPOSG *with two physical actions, a fixed dominance scheme, and a visible-context map χ such that*

$$\inf_{\pi \in \Pi_{\mathrm{erase}}} \mathrm{Reg}_T(\pi) \geq T/2,$$

while an observer-aware policy $\pi^\star$ achieves $\mathrm{Reg}_T(\pi^\star) = 0$.

Proof. Consider a repeated one-step problem with actions $A^p = \{\texttt{go}, \texttt{hold}\}$. Let two physical states x^+ and x^- have the same visible context, $\chi(x^+) = \chi(x^-) = y_0$. Let ℓ_H and ℓ_L be observer–situation locations. Let q_r be the proposition family for berth readiness, and fix the dominance scheme so that any readiness-family claim at ℓ_H strictly dominates any readiness-family claim at ℓ_L.

The two possible claim bases are

$$C^+ = \{(\mathit{ready}, \ell_H, 1, 0, \rho_1), (\neg \mathit{ready}, \ell_L, 1, 0, \rho_2)\},$$

$$C^- = \{(ready, \ell_L, 1, 0, \rho_1), (\neg ready, \ell_H, 1, 0, \rho_2)\}.$$

They have the same provenance-erased aggregate. In state $s^+ = (x^+, C^+)$, reward is one for `go` and zero for `hold`. In state $s^- = (x^-, C^-)$, reward is one for `hold` and zero for `go`. The initial state in each round is uniform over s^+ and s^-.

Any $\pi \in \Pi_{\text{erase}}$ must use the same action distribution in both states. If it selects `go` with probability p, its expected per-round reward is $\frac{1}{2}p + \frac{1}{2}(1-p) = \frac{1}{2}$. The observer-aware policy that follows the undefeated readiness-family claim under $\mathcal{D}$ obtains reward one in both states. Hence the per-round regret of every provenance-erased policy is at least $1/2$, giving cumulative regret at least $T/2$. □

Corollary 7.4 (Safety-budget separation)**.** *There exists a safety-constrained version of the construction in definition 7.3 in which any provenance-erased policy either incurs linear safety violation or linear service regret, while an observer-aware policy incurs neither.*

Proof. Set `go` in state s^- to create one unit of safety violation and set `hold` in state s^+ to create one unit of service regret. Since any provenance-erased policy uses the same action distribution in both states, decreasing one burden increases the other. The observer-aware policy conditions on the undefeated readiness-family claim and selects the correct action in both cases. □

7.3 Certification, shields, and soundness

The admissibility relation resembles a shield, but the shield condition is constructed from dominant support certificates rather than from a state predicate alone. This distinction matters for the paper's novelty boundary. If the certificate is erased and only a final binary safety label remains, the process becomes closer to standard shielding [9, 12]. If the certificate remains, the action is justified by a specific undefeated support set.

Theorem 7.5 (Certificate soundness)**.** *Fix time t, agent i, and critical physical action a_i^p. If $S \in \mathsf{DCert}_i(x_t, C_t, a_i^p)$, then there exists a classical interpretation I such that*

$$I \models \mathrm{Th}_{\lambda_i(x_t)}(S) \quad \textit{and hence} \quad I \models \mathrm{Pre}_i(a_i^p).$$

Thus the certificate is a semantically sound explanation witness for the action preconditions at the decision node.

Proof. By definition of dominant support certificate, $\mathsf{Cons}(\mathrm{Th}_{\lambda_i(x_t)}(S)) = 1$ and $\mathrm{Th}_{\lambda_i(x_t)}(S)$ entails every formula in $\mathrm{Pre}_i(a_i^p)$. Soundness of Cons gives a classical interpretation satisfying the supported theory; entailment then gives satisfaction of each action precondition. □

Proposition 7.6 (Gate safety by construction)**.** *Suppose all critical unsafe admissions are characterized by failure of at least one precondition in $\mathrm{Pre}_i(\texttt{admit})$, and suppose the dominance scheme is correct on the active proposition family in the sense that the undefeated support set entails a false precondition only when the corresponding authoritative claim is itself false. Then any policy that admits only when $\mathsf{DCert}_i(x, C, \texttt{admit}) \neq \varnothing$ and $\kappa_i(x, C, \texttt{admit}) \leq \eta_i$ has zero certificate-induced safety violations on episodes where the authoritative active claims are physically correct.*

Proof. Under the policy condition, every admission has a dominant certificate. By definition 7.5, the certificate entails the admission preconditions. By the correctness assumption for active authoritative claims, those preconditions hold physically on the considered episodes. Therefore the admission is not unsafe on those episodes. Any remaining safety violation would arise from false authoritative claims or incorrect dominance calibration, which the proposition explicitly excludes. □

Remark 7.7. Definition 7.6 is not a deployment guarantee. It states the exact assumption under which the gate is safe by construction. This is useful because it exposes what the experiment must measure: whether false authoritative claims, stale dominance, or hidden leakage occur often enough to invalidate the active regime.

7.4 What the theorems do and do not establish

Table 7 summarizes the role of each theorem. The formal results establish process-class separation and certificate soundness, not empirical superiority in a full maritime twin.

Table 7: Theory status and validation responsibility.

Result	Established by the manuscript	Left to validation or deployment calibration
Containment	OS-CPOSG specializes to standard constrained POSG/Dec-POMDP variants when claim semantics are removed.	Whether a particular maritime twin needs the richer state.
Observer-collapse separation	There exist controlled cases where provenance erasure causes linear regret under identical visible context.	Frequency and operational importance of such cases in real replay.
Safety-budget separation	Provenance erasure can force a safety–service tradeoff that observer-aware semantics avoid.	Magnitude of safety, emissions, and service burdens under the twin's metric definitions.
Certificate soundness	Certified support sets are internally satisfiable and entail action preconditions.	Whether the underlying sensors, authority hierarchy, and simulator are externally valid.
Gate safety by construction	If active authoritative claims are physically correct, certificate-gated admission prevents certificate-induced unsafe admission.	Estimating false-authority and stale-dominance rates under replay.

8 Minimal benchmark: Berth-1-Conflict

The benchmark tests whether the semantic distinction in definition 7.3 survives executable conditions. It is not intended to show that a full port is solved. It is a unit test for the process-class claim.

8.1 Environment

Each episode concerns a single berth admission decision repeated across seeds and time steps. The latent state contains four safety-relevant booleans:

$$x_t = (ready_t, craneOK_t, etaOnTime_t, weatherSafe_t, z_t),$$

where z_t contains visible non-safety context. The controller-visible map χ_{B1} exposes only z_t:

$$\chi_{B1}(x_t) = (queue\_state, weather\_regime, disruption\_family, vessel\_class, operating\_mode, time\_bucket, ber$$

It deliberately excludes the four hidden booleans and all claim provenance.

The physical and epistemic action sets are

$$A^p = \{\texttt{admit}, \texttt{hold}\}, \qquad A^e = \{\texttt{none}, \texttt{verify}, \texttt{defer}, \texttt{escalate}, \texttt{certify}\}.$$

The admission precondition family is

$$\mathrm{Pre}(\texttt{admit}) = \{ready, craneOK, etaOnTime, weatherSafe\}.$$

An unsafe admission occurs if `admit` is chosen while any of these latent preconditions is false. A service burden occurs if `hold` is chosen while all latent preconditions are true. Epistemic exception actions incur intervention cost.

8.2 Scenario families

The benchmark contains four active conflict families and one inactive control family:

1. **Authority conflict**: a high-priority local observer and a lower-priority planning observer disagree on readiness.
2. **Freshness conflict**: a fresh and stale report disagree within the same proposition family.
3. **Context conflict**: a current-situation report conflicts with a planning-situation report.
4. **Human override**: a human or institutional authority conflicts with an automated feed.
5. **Benign agreement**: observers agree; provenance semantics should introduce little or no overhead.

These families map directly to maritime sources: harbor master or terminal operations for readiness, crane maintenance for crane availability, AIS or carrier schedule for ETA, and local sensor or approved weather feed for weather admissibility. The point is not that these are the only maritime sources. The point is that the family labels are fixed before results and reused across ablations.

8.3 Ablation contract

All experimental conditions use the same latent episode bank and the same solver. The only manipulated object is the claim-state semantics:

1. **Provenance-preserving**: full observer–situation labels, pre-registered dominance, contradiction mass, and certification gates.
2. **Provenance-erased**: labels and provenance removed before decision.
3. **Random-label placebo**: labels randomized independently of authority; tests whether arbitrary metadata helps.

4. **Labels without gating**: labels visible, but certification/admissibility gates disabled; isolates provenance visibility from feasibility semantics.
5. **Oracle visible-state**: hidden readiness and related latent booleans exposed; sanity upper bound and leakage diagnostic.
6. **Benign agreement**: no relevant contradictions; tests overhead when semantics are inactive.

A positive benchmark result has a specific shape. Provenance-preserving should approach the oracle in active conflict regimes. Provenance-erased should incur paired regret or a safety–service tradeoff. Random-label placebo should not close the gap. Labels-without-gating should reveal why visibility alone is insufficient. Benign agreement should show low overhead. If this pattern does not appear, the correct response is not to add controller depth; it is to narrow the active regime or revise the dominance assumptions.

8.4 Metrics

The primary metrics are:

$$
\begin{aligned}
\mathrm{Regret}_T &= \sum_{t=0}^{T-1} (r_t^\star - r_t), \\
\mathrm{SafetyViol}_T &= \sum_{t=0}^{T-1} \mathbf{1}[\texttt{admit}_t \text{ and some latent precondition is false}], \\
\mathrm{ServiceDelay}_T &= \sum_{t=0}^{T-1} \mathbf{1}[\texttt{hold}_t \text{ and all latent preconditions are true}], \\
\mathrm{InterventionCost}_T &= \sum_{t=0}^{T-1} c_e(a_t^e).
\end{aligned}
$$

Contradiction half-life is a secondary semantic metric: the number of steps required for a contradiction component to be resolved, bypassed, or rendered irrelevant to the action. Emissions and fairness metrics are tertiary translations. An unnecessary hold can be mapped to idling emissions through a simulator-calibrated idling constant; service disparity can be measured through wait-time or cost differences across vessel or stakeholder classes. Ratio metrics such as Gini should be reported only with denominator diagnostics; absolute pairwise disparity is the preferred denominator-free companion.

9 Replay calibration and locked validation protocol

A professional validation plan separates semantic separation, replay transfer, and metric lift. This section is intentionally prescriptive because the benchmark is easy to invalidate accidentally.

9.1 Layer 0: preflight

Preflight validates the contract and writes no result CSVs. It checks the visible-context leakage guard, condition set, dominance-table version, solver version, replay schema, known observers, known proposition families, and manifest path. Preflight may be run on the synthetic bank or on a proposed replay CSV. It is not evidence.

9.2 Layer 1: semantic separation

Run Berth-1-Conflict with paired seeds and a frozen solver. The required result is a paired difference between provenance-preserving and provenance-erased conditions on zero-violation regret or the safety–service tradeoff. The random-label placebo and benign-agreement conditions are essential controls: without them, a positive result could be attributed to arbitrary labels or to generic conservatism.

9.3 Layer 2: maritime replay

Replace synthetic claims with a replay CSV exported from the maritime twin. The replay schema must include episode identifier, step, visible-context features, latent readiness or admissibility label if available for evaluation, observer, situation, assertion, timestamp, credibility, provenance, and intervention outcome. The solver and ablation interface remain unchanged. A replay path must be an actual file; placeholder paths such as `path/to/twin_replay_claims.csv` are documentation only.

9.4 Layer 3: metric lift

Only after Layer 1 separates should the analysis translate burdens into CH-MARL-style metrics: idling emissions, service disparity, intervention spend, and fairness observables. Confidence statements should use block-level replay units keyed by traffic family, weather regime, disruption family, horizon, operating mode, and simulator version. Hoeffding-style bounds or empirical distribution bounds can be used for one-sided coverage when the block assumptions are appropriate [49, 50].

Table 8: Experimental lock items. These are fixed before local experiments.

Item	Required lock
Visible context	Define $\chi_{B1}(x)$ and verify it excludes latent safety truth and claim provenance.
Dominance table	Pre-register observer priority by proposition family; do not tune after seeing results.
Freshness windows	Fix timestamp decay and stale-claim thresholds by proposition family.
Ablation set	Use the six conditions in §8; no solver changes across conditions.
Episode bank	Generate or export latent episodes once; reuse across all conditions.
Metrics	Primary: zero-violation regret and safety–service tradeoff. Secondary: emissions/fairness translations.
Stopping rule	Scale to full twin only if fixed-solver provenance preservation separates from erasure and placebo.
Shareable output	Summary CSV, paired-delta CSV, diagnostics JSON, manifest JSON, preflight manifest, and short report.

10 Implementation witness

The artifact package contains a `Berth-1-Conflict` scaffold. Its role is contractual rather than evidentiary. In preflight mode it validates schema, manifests, ablation condition names, and output paths without generating results. Result generation is intentionally explicit. This prevents accidental synthetic smoke tests from being treated as evidence.

Table 9: Executable modes and evidentiary status.

Mode	Command intent	Evidentiary status
Preflight	Validate schema, leakage guard, manifest, condition names, and dominance versions.	Contract check only. No results.
Synthetic smoke	Run the fixed solver on generated episodes with explicit `-allow-synthetic-results`.	Software sanity check. Not maritime evidence unless declared as a locked synthetic benchmark.
Replay run	Run the same ablation interface on an actual replay CSV exported from the maritime twin.	First admissible empirical evidence for the paper.
Packaging	Bundle summary, paired deltas, diagnostics, manifests, and report.	Shareable artifact for review and follow-up analysis.

The dominance-gated EcoFair-style wrapper is an implementation witness, not the contribution. Any MARL solver can be used if it obeys the frozen-solver ablation contract. This interpretation is supported only if the separation tracks the process-state semantics rather than the solver identity. In particular, local experiments should first run the scaffold exactly as specified; solver improvements belong in a later study after semantic separation is established.

11 Limitations and threats to validity

The central limitation is scope. The separation theorem proves that observer–situation provenance can be necessary, not that every maritime coordination instance requires OS-CMASP semantics. The active regime is expected to be situations with safety-critical or service-critical actions, competing reports, and pre-registered authority/freshness rules. In low-conflict or purely stochastic-noise regimes, a fused belief state may be sufficient.

A second limitation is authority calibration. Dominance orders are domain artifacts. If authority or freshness rules are tuned after observing benchmark outcomes, the experiment becomes invalid. This is why the pre-run lock treats $\mathcal{D}$, freshness windows, and tie rules as versioned process parameters.

A third limitation is the distinction between logical soundness and empirical truth. Certificate soundness means that the emitted support set is internally satisfiable and entails the stated action preconditions. It does not prove that the world satisfies those preconditions. Physical validity still depends on calibrated sensors, reliable human authority, and simulator fidelity.

A fourth limitation concerns secondary metrics. Fairness and emissions translations are meaningful only when replay calibration supports them. If denominator diagnostics fail for Gini, Gini should be reported as descriptive only, while direct service proxies or absolute pairwise disparity carry the proof burden.

Table 10: Main threats to validity and protocol controls.

Threat	Control
Hidden truth leaks through ordinary features	Lock χ_{B1}; run a leakage test on the provenance-erased condition.
Dominance order is outcome-tuned	Register $\mathcal{D}$, freshness windows, and tie rules before replay.
Solver improvement is mistaken for semantics	Freeze solver, seeds, action costs, and latent episode bank.
Synthetic smoke tests are mistaken for evidence	Require an explicit synthetic flag and label outputs as software checks.
Fairness lift overclaims beyond replay support	Certify only positive replay coefficients; otherwise report as diagnostics.
Minimal benchmark lacks ecological validity	Use only as an active-regime screen before maritime-twin replay.

12 Conclusion

OS-CMASP proposes a process-level account of regulated multi-agent coordination under fragmented operational truth. Its novelty is not a new hierarchy, belief layer, or LLM wrapper. It is the decision-theoretic status of observer–situation provenance: when contradictions are action-relevant, erasing provenance can alter regret and feasibility even under a frozen solver. The formal results establish that provenance can be part of the control problem rather than only part of an explanation layer.

The next empirical step is a locked, paired, solver-isolated BERTH-1-CONFLICT run. The first local run should be preflight. The second should be a synthetic smoke test only to verify the pipeline. The first paper-relevant run should use an actual replay CSV exported from the maritime twin. If the minimal separation does not appear, the response should not be to add controller depth; it should be to identify the narrower operational regime in which observer–situation provenance is active.

A Proof details for the safety-budget construction

Let p be the probability that a provenance-erased policy chooses `go`. In state s^-, `go` creates one safety violation, so expected safety violation per round is $p/2$. In state s^+, `hold` creates one service-regret unit, so expected service regret per round is $(1-p)/2$. Thus for any p, at least one of safety violation or service regret is at least $1/4$ per round, and their sum is exactly $1/2$ per round. The observer-aware policy chooses `go` in s^+ and `hold` in s^-, so both quantities are zero.

B Replay schema

A minimal replay CSV should contain the following columns:

Column	Meaning
episode_id	Stable episode identifier.
step	Integer decision step.
context_json	Serialized visible context used by χ_{B1}.
latent_ready	Evaluation-only readiness label, if available.
observer	Observer or information source.
situation	Situation identifier or regime label.
claim	Formula or literal such as `ready` or `not_ready`.
credibility	Numeric confidence or calibrated source reliability.
timestamp	Event or assertion timestamp.
provenance	Source metadata required for audit.
outcome_json	Optional action/outcome fields from the twin.

C Bibliographic positioning

The bibliography distinguishes three classes of citation. Foundational sources establish stochastic games, Dec-POMDPs, constrained control, epistemic reasoning, truth maintenance, and logistics. Contemporary preprints establish immediate technical neighbors: OSL, EcoFair-CH-MARL, CH-MARL, and R-CMASP. Project websites, newsroom announcements, and internal deployment material do not carry technical claims in the paper. They may motivate case-study selection in a cover note, acknowledgments, or a non-archival deployment memo, but they do not substitute for scholarly references.

D Technical supplement: budget lifts, replay certification, and operationalization

This appendix collects the extended theory, validation machinery, and operationalization details that support the formal core. The claims here are subordinate to the main process-semantics claim: they specify how to test, calibrate, and operationalize OS-CMASP without turning the paper into a controller-stack contribution.

E Extended non-stacking rationale

The extended rationale is organized around four design decisions.

1. **State semantics.** The environment state is not only a latent physical state x_t but also a claim base C_t over observer–situation space. Claims are not immediately collapsed into a single fused estimate.

2. **Action semantics.** Joint action includes both physical control acts and epistemic repair acts such as query, verify, defer, challenge, certify, and escalate.
3. **Feasibility semantics.** A critical action is admissible only if its preconditions are support-certified under a satisfiable theory or routed through an explicit epistemic exception act.
4. **Audit semantics.** Explanations are not post-hoc summaries of hidden computations. They are support certificates drawn from the same contradiction-aware claim state that the controller uses at decision time.

E.1 Non-stacking criterion and refinement ladder

To keep the novelty claim disciplined, it helps to say explicitly what *would not* count as a new contribution here.

Definition E.1 (Stacked architectural augmentation)**.** Fix a base environment class

$$\mathcal{G}_0 = \langle X, \{A_i^p\}_{i\in N}, P_0, \mathcal{F}_0, J_0\rangle$$

with latent state space X, physical action interface $\{A_i^p\}$, transition law P_0, admissibility relation $\mathcal{F}_0$, and objective / constraint bundle J_0. A proposal is a *stacked architectural augmentation* of $\mathcal{G}_0$ iff it preserves the environment-level tuple

$$(X, \{A_i^p\}_{i\in N}, P_0, \mathcal{F}_0, J_0)$$

up to observationally irrelevant reparameterization, and changes only the controller internals: policy class, auxiliary losses, memory structure, communication architecture, or post-hoc explanation module.

Remark E.2 (What this definition rules out)*.* By Definition E.1, proposals of the form "EcoFair-CH-MARL plus an OSL module" are still stacked unless OSL changes what the environment state *is*, what actions *exist*, or what makes a critical action *admissible*. Better performance alone would not justify a process-class claim.

Under that criterion, process-class novelty is warranted only to the extent that all three of the following are true.

R1: **State test.** Preserving observer–situation provenance changes the decision problem, not merely the quality of an internal estimate.

R2: **Feasibility test.** Certification / escalation changes which actions are executable, not merely which explanations can be shown after execution.

R3: **Solver-independence test.** The core gain survives a fixed-controller ablation in which only state semantics and admissibility semantics are changed.

These rejection conditions are the reason the separation theorem in Section 7 matters more than any specific neural architecture.

E.2 Cautionary scope and non-claims

Because the paper is trying to move from *architecture* novelty to *problem-class* novelty, it is important to say explicitly what is *not* being claimed.

N1: Not the first epistemic MARL model. Interactive belief-state methods, common-knowledge coordination, and epistemic planning already show that epistemic structure can be control-relevant [19, 26, 28].

N2: Not the first simulator-coupled governed process. R-CMASP already provides a close process-level neighbor with simulator-coupled dynamics and normative feasibility [43].

N3: Not a theorem about the full maritime twin. The separation results in Section 7 are intentionally minimal. They establish the candidate distinction on berth-admission-style families, not yet on the full 16-port digital twin.

N4: Not a claim that one dominance order is canonical. The priority map π_L is a modeling choice. Different ports may instantiate authority, recency, and source trust differently.

The guarded claim is therefore narrower but not extension-shaped. If the separation theorems survive and the fixed-solver ablations in Section I confirm solver-independence, then OS-CMASP is a defensible *candidate* process class for fragmented-truth control. If a specific empirical route fails, the appropriate refinement is to restrict the active regime to settings where provenance affects admissibility, contradiction burden, or replay-certified budget metrics under a fixed solver. The fallback is a sharper OS-CMASP subcase, not an auxiliary module attached to an existing controller.

These choices position the model relative to three adjacent lines of work. First, interactive and belief-state models such as I-POMDPs enrich the state space with models of other agents, but they still treat the problem as planning over a belief state rather than over an explicit contradiction-bearing claim base [28]. Second, common-knowledge methods such as MACKRL exploit information the agents can agree on without communication; this is powerful for decentralized coordination, but it presumes a form of agreement rather than making disagreement a first-class state object [19]. Third, epistemic planners with common knowledge focus on actions with epistemic preconditions and effects, but they are not formulated as simulator-coupled constrained control problems with cumulative operational objectives and feasibility gating [26]. The closest neighboring formalism is therefore R-CMASP, not the broader epistemic-planning literature.

F Expanded budget-bundle and replay-calibration machinery

F.1 A CH-MARL-shaped budget bundle

Definition F.1 (Minimal CH-MARL-style budget bundle)**.** Consider a berth-admission benchmark with vessel class label

$$c_t \in \{A, B\}$$

and binary latent readiness variable

$$r_t \in \{0, 1\}.$$

Define the cumulative safety burden by

$$V_T(\pi) := \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \mathbf{1}[\mathcal{F}(x_t, C_t, a_t) = 0]\right].$$

For each vessel class $c \in \{A, B\}$, define the unnecessary ready-hold count

$$U_c(T;\pi) := \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \mathbf{1}[c_t = c,\ r_t = 1,\ a_t^p = \mathsf{hold}]\right].$$

Define idling-emission burden, service-disparity burden, and intervention spend by

$$E_T(\pi) := U_A(T;\pi) + U_B(T;\pi),$$

$$F_T(\pi) := \big|U_A(T;\pi) - U_B(T;\pi)\big|,$$

$$I_T(\pi) := \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \mathbf{1}[a_t^e \neq \varnothing]\right].$$

We call

$$\mathcal{B}_T(\pi) := (V_T(\pi), E_T(\pi), F_T(\pi), I_T(\pi))$$

the minimal safety–emission–fairness–intervention bundle.

Remark F.2 (Why this fairness metric is intentionally minimal). The quantity F_T is not intended to reproduce the full fairness machinery of EcoFair-CH-MARL. It is a benchmark-side service-parity proxy chosen for one reason only: it exposes a provenance-sensitive delay asymmetry in a form that can be analyzed without changing the solver. If this proxy fails to separate the provenance-preserving and provenance-erased problems, the stronger fairness story is withheld.

Theorem F.3 (Two-class bundle impossibility under provenance erasure)**.** *For every even horizon $T \geq 2$, there exists a two-class berth-admission family of* OS-CPOSG *instances with the following properties.*

(i) On every class-A round, the berth is ready and the claim base admits an unambiguous dominant certificate for ready.

(ii) On every class-B round, the claim base is either C_B^+ or C_B^- with equal probability, where

$$C_B^+ = \{(\mathit{ready}, \ell_H, 1, 0, \rho_1), (\neg\mathit{ready}, \ell_G, 1, 0, \rho_2)\},$$

$$C_B^- = \{(\mathit{ready}, \ell_G, 1, 0, \rho_1), (\neg\mathit{ready}, \ell_H, 1, 0, \rho_2)\},$$

with $\pi_L(\ell_H) > \pi_L(\ell_G)$ and

$$\mathrm{Agg}(C_B^+) = \mathrm{Agg}(C_B^-).$$

(iii) The reward is 1 *for the correct physical action on each round and* 0 *otherwise; on class-A rounds, the correct action is* admit*, while on class-B rounds the correct action is the action supported by the dominant claim.*

(iv) The action admit *is critical, and on class-B rounds it is admissible iff a dominant support certificate for ready exists.*

Then there exists an observer-aware policy $\pi^\star$ *such that*

$$\mathcal{B}_T(\pi^\star) = (0,0,0,0) \qquad \textit{and} \qquad J_T(\pi^\star) = T,$$

whereas no aggregated policy $\pi \in \Pi_{\text{agg}}$ *can satisfy*

$$\mathcal{B}_T(\pi) = (0,0,0,0).$$

Proof. Consider any aggregated policy $\pi \in \Pi_{\text{agg}}$. Because class-A rounds are unambiguous, let

$$q := \pi(\mathsf{admit} \mid \mathrm{Agg}(C_A))$$

denote the admission probability on class-A rounds. Because

$$\mathrm{Agg}(C_B^+) = \mathrm{Agg}(C_B^-),$$

there is a single admission probability

$$p := \pi(\mathsf{admit} \mid \mathrm{Agg}(C_B^+)) = \pi(\mathsf{admit} \mid \mathrm{Agg}(C_B^-))$$

for all class-B rounds without extra epistemic resolution.

There are $T/2$ class-A rounds and $T/2$ class-B rounds. On class-A rounds, every hold is an unnecessary ready-hold, so

$$U_A(T;\pi) = \frac{(1-q)T}{2}.$$

On class-B rounds, the berth is ready in exactly half the rounds in expectation, and holding on those ready rounds creates unnecessary delay and idling, so

$$U_B(T;\pi) = \frac{(1-p)T}{4}.$$

Likewise, admitting on the not-ready class-B rounds is inadmissible, so

$$V_T(\pi) = \frac{pT}{4}.$$

Since $E_T = U_A + U_B$ and $F_T = |U_A - U_B|$, the zero-bundle equalities

$$V_T(\pi) = E_T(\pi) = F_T(\pi) = I_T(\pi) = 0$$

would imply, in particular, $V_T(\pi) = 0$ and $E_T(\pi) = 0$. The first forces $p = 0$. The second then forces $U_A = U_B = 0$, hence $q = 1$ and simultaneously $U_B = (1-p)T/4 = T/4$, a contradiction. Therefore no aggregated policy can satisfy the zero bundle.

Now define the observer-aware policy $\pi^\star$ by: admit on every class-A round; on class-B rounds, admit in state C_B^+ and hold in state C_B^-. This policy is always admissible, never holds a ready vessel unnecessarily, never uses an epistemic exception act, and chooses the correct action on every round. Hence

$$\mathcal{B}_T(\pi^\star) = (0,0,0,0) \qquad \text{and} \qquad J_T(\pi^\star) = T.$$

$\square$

Corollary F.4 (Linear budget tradeoff under provenance erasure). *In the family of Theorem F.3, every aggregated policy $\pi \in \Pi_{\text{agg}}$ satisfies*

$$V_T(\pi) + \frac{1}{2}\big(E_T(\pi) + F_T(\pi)\big) \geq \frac{T}{4}.$$

Consequently, no provenance-erased aggregated policy can drive safety burden, idling-emission burden, and service-disparity burden all to $o(T)$ simultaneously.

Proof. From the proof of Theorem F.3,

$$V_T(\pi) = \frac{pT}{4}.$$

Moreover,

$$E_T(\pi) + F_T(\pi) = U_A + U_B + |U_A - U_B| = 2\max\{U_A, U_B\} \geq 2U_B = \frac{(1-p)T}{2}.$$

Therefore

$$V_T(\pi) + \frac{1}{2}\big(E_T(\pi) + F_T(\pi)\big) \geq \frac{pT}{4} + \frac{(1-p)T}{4} = \frac{T}{4}.$$

□

Corollary F.5 (Intervention can only buy down the burden linearly). *Extend the family of Theorem F.3 by allowing an epistemic action* resolve *on class-B rounds. Each use of* resolve *reveals whether the current class-B round is in state C_B^+ or C_B^- before the physical action is chosen, and each use contributes one unit to I_T. Then every aggregated policy satisfies*

$$V_T(\pi) + \frac{1}{2}\big(E_T(\pi) + F_T(\pi)\big) + \frac{1}{2}I_T(\pi) \geq \frac{T}{4}.$$

Hence a provenance-erased controller can remove the linear safety / emission / fairness burden only by paying linear intervention spend.

Proof. Let $N_B = T/2$ be the number of class-B rounds and let $M_T \leq I_T(\pi)$ denote the expected number of class-B rounds on which the policy invokes resolve. On the remaining unresolved class-B rounds, the policy still sees only the collapsed view $\text{Agg}(C_B^+) = \text{Agg}(C_B^-)$, so the same argument as in Corollary F.4 applies on that unresolved subset. Therefore

$$V_T(\pi) + \frac{1}{2}\big(E_T(\pi) + F_T(\pi)\big) \geq \frac{N_B - M_T}{2} = \frac{T}{4} - \frac{M_T}{2}.$$

Since $M_T \leq I_T(\pi)$,

$$V_T(\pi) + \frac{1}{2}\big(E_T(\pi) + F_T(\pi)\big) + \frac{1}{2}I_T(\pi) \geq \frac{T}{4}.$$

□

Remark F.6 (What the bundle theorem does and does not establish). Theorem F.3 and Corollaries F.4–F.5 are not claims about the full EcoFair digital twin or its full fairness transformer. They show a narrower fact that is directly useful for the novelty test: even in a fixed-solver berth-admission family, provenance erasure changes the feasible budget geometry. This is a process-semantics result, not a stacked-controller result.

F.2 Toward an EcoFair-aligned budget lift

The minimal bundle above is intentionally benchmark-native. To connect it more cautiously to EcoFair-style accounting without claiming that the full digital twin has already been analyzed, we introduce actual budget-facing metrics and state the weakest calibration assumptions under which the bundle separation survives.

Definition F.7 (EcoFair-aligned benchmark metrics)**.** Fix budgets $B_E, B_F, B_I \geq 0$. Let

$$\mathcal{E}_T^{\mathrm{idle}}(\pi)$$

denote cumulative extra idling-emission units relative to immediate certified admission, let

$$\mathcal{D}_T(\pi)$$

denote a chosen service-disparity statistic for the experiment (for example, the absolute difference in class-wise mean ready-to-berth delay), and let

$$\mathcal{X}_T^E(\pi) := (\mathcal{E}_T^{\mathrm{idle}}(\pi) - B_E)_+, \qquad \mathcal{X}_T^F(\pi) := (\mathcal{D}_T(\pi) - B_F)_+, \qquad \mathcal{X}_T^I(\pi) := (I_T(\pi) - B_I)_+.$$

We refer to $(V_T, \mathcal{X}_T^E, \mathcal{X}_T^F, \mathcal{X}_T^I)$ as the EcoFair-aligned safety–emission–fairness–intervention overshoot vector.

Definition F.8 (Benchmark-side calibration assumptions)**.** We say that a berth-admission benchmark family admits an $(\underline{\epsilon}, \underline{\phi})$-calibration if there exist constants $\underline{\epsilon}, \underline{\phi} > 0$ such that for every policy π in the family,

$$\mathcal{E}_T^{\mathrm{idle}}(\pi) \geq \underline{\epsilon}\, E_T(\pi) \qquad \text{and} \qquad \mathcal{D}_T(\pi) \geq \underline{\phi}\, F_T(\pi).$$

The first inequality says that every unnecessary ready-hold induces at least $\underline{\epsilon}$ units of extra idling emissions on the benchmark. The second says that the selected fairness statistic is sensitive enough to lower-bound the class-asymmetry exposed by F_T.

Proposition F.9 (EcoFair-aligned metric lift)**.** *Assume the two-class berth-admission family of Theorem F.3 admits an $(\underline{\epsilon}, \underline{\phi})$-calibration in the sense of Definition F.8. Then every aggregated policy $\pi \in \Pi_{\mathrm{agg}}$ satisfies*

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}}\, \mathcal{E}_T^{\mathrm{idle}}(\pi) + \frac{1}{2\underline{\phi}}\, \mathcal{D}_T(\pi) + \frac{1}{2} I_T(\pi) \geq \frac{T}{4}.$$

Hence provenance erasure cannot be hidden by merely swapping in more realistic emission and service metrics, provided those metrics are benchmark-calibrated to the underlying ready-hold geometry.

Proof. By Definition F.8,

$$\frac{1}{2\underline{\epsilon}}\, \mathcal{E}_T^{\mathrm{idle}}(\pi) \geq \frac{1}{2} E_T(\pi) \qquad \text{and} \qquad \frac{1}{2\underline{\phi}}\, \mathcal{D}_T(\pi) \geq \frac{1}{2} F_T(\pi).$$

Substituting these bounds into Corollary F.5 yields

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}}\, \mathcal{E}_T^{\mathrm{idle}}(\pi) + \frac{1}{2\underline{\phi}}\, \mathcal{D}_T(\pi) + \frac{1}{2} I_T(\pi) \geq \frac{T}{4}.$$

□

Corollary F.10 (EcoFair-aligned overshoot tradeoff)**.** *Under the assumptions of Proposition F.9, every aggregated policy $\pi \in \Pi_{\text{agg}}$ satisfies*

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}}\,\mathcal{X}_T^E(\pi) + \frac{1}{2\underline{\phi}}\,\mathcal{X}_T^F(\pi) + \frac{1}{2}\,\mathcal{X}_T^I(\pi) \geq \frac{T}{4} - \frac{B_E}{2\underline{\epsilon}} - \frac{B_F}{2\underline{\phi}} - \frac{B_I}{2}.$$

In particular, if $B_E = o(T)$, $B_F = o(T)$, and $B_I = o(T)$, then provenance-erased aggregated control cannot keep all four coordinates of the overshoot vector sublinear.

Proof. From Proposition F.9,

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}}\,\mathcal{E}_T^{\text{idle}}(\pi) + \frac{1}{2\underline{\phi}}\,\mathcal{D}_T(\pi) + \frac{1}{2}I_T(\pi) \geq \frac{T}{4}.$$

Using

$$\mathcal{E}_T^{\text{idle}}(\pi) \leq \mathcal{X}_T^E(\pi) + B_E, \qquad \mathcal{D}_T(\pi) \leq \mathcal{X}_T^F(\pi) + B_F, \qquad I_T(\pi) \leq \mathcal{X}_T^I(\pi) + B_I,$$

we obtain

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}}\,\mathcal{X}_T^E(\pi) + \frac{1}{2\underline{\phi}}\,\mathcal{X}_T^F(\pi) + \frac{1}{2}\,\mathcal{X}_T^I(\pi) \geq \frac{T}{4} - \frac{B_E}{2\underline{\epsilon}} - \frac{B_F}{2\underline{\phi}} - \frac{B_I}{2}.$$

If all budgets are $o(T)$, the right-hand side remains $\Omega(T)$, so at least one coordinate of the overshoot vector must be linear in T. □

Remark F.11 (Why this lift is still cautious)*.* Proposition F.9 and Corollary F.10 do not claim that the EcoFair fairness transformer itself has already been reproduced. They only say that if the chosen benchmark-side emission and disparity metrics are calibrated to the delay geometry created by unnecessary ready-holds, then the minimal bundle impossibility propagates to a more CH-MARL-shaped budget language without changing the solver identity.

F.3 Simulator-grounded calibration by counterfactual replay

EcoFair's maritime digital twin already exposes the ingredients needed to replace abstract calibration rhetoric with measured quantities: hourly decision epochs, explicit queueing and idling fuel consumption when berth capacity binds, and per-vessel cost logs from which fairness diagnostics such as Gini and min–max are computed [14]. The least risky next step is therefore not to add more controller depth, but to use the twin itself as a measurement device while keeping the solver frozen.

Definition F.12 (Twin-native idling constant)**.** Fix a berth-admission witness slice

$$\mathcal{W}_{\text{ready}} \subseteq X \times 2^{\text{Lit}(\Phi)\times L\times[0,1]\times\mathbb{T}\times\mathcal{P}}$$

consisting of states (x_t, C_t) in which a designated vessel is physically berth-ready, immediate certified admit is feasible, and a physical hold leaves that vessel idling for one simulator step of length $\Delta t > 0$. Let

$$\dot{e}_{\text{idle}}(x_t)$$

be the simulator's idling-emission rate for that vessel at state x_t. Define

$$\underline{\epsilon}_{\mathcal{S}} := \inf_{(x_t,C_t)\in\mathcal{W}_{\text{ready}}} \dot{e}_{\text{idle}}(x_t)\,\Delta t.$$

We call $\underline{\epsilon}_{\mathcal{S}}$ the twin-native idling constant.

Proposition F.13 (Emission calibration from twin-native idling semantics)**.** *Assume the berth-admission benchmark uses the maritime digital twin semantics of EcoFair-CH-MARL, with explicit idling-emission accumulation during queueing and a decision step of length Δt. If every unnecessary ready-hold counted by $E_T(\pi)$ occurs at some state in $\mathcal{W}_{\text{ready}}$, then every policy π satisfies*

$$\mathcal{E}_T^{\text{idle}}(\pi) \geq \underline{\epsilon}_{\mathcal{S}}\, E_T(\pi).$$

Hence the emission-side calibration constant in Definition F.8 can be instantiated directly from the simulator semantics rather than postulated abstractly.

Proof. Each unnecessary ready-hold contributes at least one simulator step of idling for the held vessel, and by Definition F.12 that step carries at least $\underline{\epsilon}_{\mathcal{S}}$ extra idling emissions. Summing this nonnegative contribution over all unnecessary ready-holds counted by $E_T(\pi)$ gives

$$\mathcal{E}_T^{\text{idle}}(\pi) \geq \underline{\epsilon}_{\mathcal{S}}\, E_T(\pi).$$

□

Definition F.14 (Episode-matched fairness calibration record)**.** Fix a frozen solver and a chosen fairness proxy $\mathcal{D}_T$ for the berth-admission benchmark. Run the simulator for episodes indexed by $m \in \{1, \dots, M\}$ under common random seeds and log the episode-level pairs

$$\mathfrak{R}_M := \{(F_T^{(m)}, \mathcal{D}_T^{(m)})\}_{m=1}^{M}.$$

A function $L_M : \mathbb{N} \to \mathbb{R}_+$ is a lower calibration envelope for $\mathfrak{R}_M$ if

$$L_M(F_T^{(m)}) \leq \mathcal{D}_T^{(m)} \qquad \text{for all } m \in \{1, \dots, M\}.$$

We say that the fairness proxy is simulator-certified at slope $\underline{\phi}_{\mathcal{S}} > 0$ on an operational range $\mathcal{R} \subseteq \mathbb{N}$ if there exists a lower calibration envelope with

$$L_M(f) \geq \underline{\phi}_{\mathcal{S}} f \qquad \text{for all } f \in \mathcal{R}.$$

In practice, L_M should be chosen as a lower confidence envelope rather than as a raw sample minimum.

Definition F.15 (Quantile-certified replay slope)**.** Consider only replay episodes with $F_T^{(m)} > 0$, and define the realized ratio

$$R^{(m)} := \frac{\mathcal{D}_T^{(m)}}{F_T^{(m)}}.$$

Assume these ratios are exchangeable draws from a replay distribution $\mathbb{P}_{\mathcal{S}}$, and let

$$R_{(1)} \leq \cdots \leq R_{(M_+)}$$

be the order statistics over the M_+ positive-F_T episodes. For a target tail level $\beta \in (0, 1)$ and confidence level $1 - \alpha \in (0, 1)$, define

$$k_{\alpha,\beta} := \min \left\{ k \in \{1, \dots, M_+\} : \sum_{j=0}^{k-1} \binom{M_+}{j} \beta^j (1-\beta)^{M_+ - j} \leq \alpha \right\}.$$

If the set above is empty, certification at the requested (α, β) pair fails on the available replay bank. Otherwise define

$$\widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}} := R_{(k_{\alpha,\beta})}.$$

We call $\widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}}$ a one-sided $(1-\alpha)$-confidence lower bound for a $(1-\beta)$-coverage fairness slope.

Proposition F.16 (Confidence-certified replay envelope)**.** *Assume Definition F.15 and suppose $k_{\alpha,\beta}$ exists. Then*

$$\mathbb{P}\Big[\mathbb{P}_{R\sim\mathbb{P}_{\mathcal{S}}}\Big(R \geq \widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}}\Big) \geq 1-\beta\Big] \geq 1-\alpha.$$

Equivalently, with confidence at least $1-\alpha$ over the sampled replay bank, at least a $1-\beta$ fraction of replay episodes from the same seed distribution satisfy

$$\mathcal{D}_T \geq \widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}}\, F_T.$$

Episodes with $F_T = 0$ satisfy the inequality trivially.

Proof. Let

$$q_\beta := \sup\{q \in \mathbb{R}_+ : \mathbb{P}_{R\sim\mathbb{P}_{\mathcal{S}}}(R \geq q) \geq 1-\beta\}.$$

Then $\mathbb{P}(R \leq q_\beta) \geq \beta$. The event $R_{(k_{\alpha,\beta})} > q_\beta$ implies that fewer than $k_{\alpha,\beta}$ sampled ratios fall below or at q_β. Since the count of such ratios is stochastically lower-bounded by a Binomial(M_+, β) random variable,

$$\mathbb{P}\Big(R_{(k_{\alpha,\beta})} > q_\beta\Big) \leq \sum_{j=0}^{k_{\alpha,\beta}-1} \binom{M_+}{j} \beta^j (1-\beta)^{M_+-j} \leq \alpha.$$

Hence, with probability at least $1-\alpha$, we have $\widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}} = R_{(k_{\alpha,\beta})} \leq q_\beta$. By definition of q_β, this implies

$$\mathbb{P}_{R\sim\mathbb{P}_{\mathcal{S}}}\Big(R \geq \widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}}\Big) \geq 1-\beta.$$

Replacing R with $\mathcal{D}_T/F_T$ yields the stated inequality on all positive-F_T episodes, while the zero-F_T episodes satisfy it automatically. □

Corollary F.17 (Confidence-certified budget lift on a replay distribution)**.** *Suppose Proposition F.13 holds with twin-native idling constant $\underline{\epsilon}_{\mathcal{S}} > 0$, and suppose Definition F.15 produces $\widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}} > 0$ for the chosen fairness proxy. Then, with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of replay episodes satisfy*

$$V_T + \frac{1}{2\underline{\epsilon}_{\mathcal{S}}}\,\mathcal{E}_T^{\mathrm{idle}} + \frac{1}{2\widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}}}\,\mathcal{D}_T + \frac{1}{2} I_T \geq \frac{T}{4}.$$

Thus a deterministic lower-envelope theorem can be replaced, on a fixed seed distribution, by a one-sided confidence statement without changing the solver identity.

Proof. By Proposition F.13, every replay episode satisfies

$$\mathcal{E}_T^{\mathrm{idle}} \geq \underline{\epsilon}_{\mathcal{S}} E_T.$$

By Proposition F.16, with confidence at least $1-\alpha$, at least a $1-\beta$ fraction of replay episodes satisfy

$$\mathcal{D}_T \geq \widehat{\phi}_{\beta,\alpha}^{\mathrm{LCB}} F_T.$$

On that same event, Corollary F.5 yields the stated inequality episode-wise for those replay episodes. □

Definition F.18 (Class-separated cost witness)**.** Fix a replay episode with vessel classes A and B, cardinalities $n_A := |A|$ and $n_B := |B|$, and realized per-vessel service-cost vector $\mathbf{c} = (c_1, \ldots, c_n) \in \mathbb{R}^n_+$ with $n = n_A + n_B$. We say that the episode admits a *class-separated cost witness* at baseline $c_0 \geq 0$ and burden slope $\underline{d} > 0$ if there exist nonnegative burden increments $(\delta_j)_{j \in B}$ such that

$$c_i = c_0 \quad (i \in A), \qquad c_j = c_0 + \delta_j \quad (j \in B),$$

and

$$\sum_{j \in B} \delta_j \geq \underline{d}\, F_T.$$

If, in addition,

$$\sum_{i=1}^{n} c_i \leq U_C$$

throughout an operational range, we say the witness is *range-bounded* by U_C.

Proposition F.19 (Min–max carries a lower envelope on class-separated witnesses)**.** *Under Definition F.18, let*

$$\mathrm{MM}_T := \max_{1 \leq i \leq n} c_i - \min_{1 \leq i \leq n} c_i.$$

Then

$$\mathrm{MM}_T \geq \frac{\underline{d}}{n_B} F_T.$$

Hence min–max is a proof-carrying fairness observable on this witness family.

Proof. Because every class-A vessel remains at baseline, we have

$$\min_i c_i = c_0.$$

Among the n_B class-B vessels, at least one attains burden at least the class-B average, so

$$\max_i c_i \geq c_0 + \frac{1}{n_B} \sum_{j \in B} \delta_j.$$

Subtracting and using Definition F.18 gives

$$\mathrm{MM}_T \geq \frac{1}{n_B} \sum_{j \in B} \delta_j \geq \frac{\underline{d}}{n_B} F_T.$$

□

Proposition F.20 (Conditional Gini lower envelope on range-bounded class-separated witnesses)**.** *Under Definition F.18, let*

$$\mathrm{Gini}_T := \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |c_i - c_j|}{2n \sum_{i=1}^{n} c_i}.$$

If the witness is range-bounded by U_C, then

$$\mathrm{Gini}_T \geq \frac{n_A \underline{d}}{n U_C} F_T.$$

Hence Gini is also proof-carrying on this witness family, but only after imposing a bounded operational range.

Proof. For every ordered pair $(i, j) \in A \times B$ we have

$$|c_i - c_j| = \delta_j,$$

and the same holds for every ordered pair $(j, i) \in B \times A$. Therefore the Gini numerator satisfies

$$\sum_{i=1}^{n} \sum_{j=1}^{n} |c_i - c_j| \geq 2n_A \sum_{j \in B} \delta_j \geq 2n_A \underline{d} F_T.$$

By range-boundedness,

$$2n \sum_{i=1}^{n} c_i \leq 2nU_C.$$

Dividing the numerator bound by the denominator bound yields

$$\mathrm{Gini}_T \geq \frac{n_A \underline{d}}{nU_C} F_T.$$

□

F.4 Relaxing witness assumptions under drift and spillover

The clean witness family above is useful for a first separation result, but it is deliberately strict: every class-A vessel stays exactly at baseline and all burden sits in class B. A stronger novelty test is to keep the solver frozen and relax those assumptions. The next definition allows the nominally unaffected class to drift upward and absorbs multi-subqueue spillover into an aggregate contamination term.

Definition F.21 (Drift-contaminated class-separated witness)**.** Fix a replay episode with vessel classes A and B, cardinalities $n_A := |A|$ and $n_B := |B|$, and realized per-vessel service-cost vector $\mathbf{c} = (c_1, \ldots, c_n) \in \mathbb{R}_+^n$ with $n = n_A + n_B$. We say that the episode admits a *drift-contaminated class-separated witness* at baseline $c_0 \geq 0$, burden slope $\underline{d} > 0$, class-A band $\overline{\Delta}_A \geq 0$, and class-$A$ drift slope $\overline{\Lambda}_A \geq 0$ if there exist nonnegative numbers $(\alpha_i)_{i \in A}$ and $(\delta_j)_{j \in B}$ such that

$$c_i = c_0 + \alpha_i \quad (i \in A), \qquad 0 \leq \alpha_i \leq \overline{\Delta}_A,$$

$$c_j \geq c_0 + \delta_j \quad (j \in B),$$

and

$$\sum_{j \in B} \delta_j \geq \underline{d}\, F_T, \qquad \sum_{i \in A} \alpha_i \leq \overline{\Lambda}_A\, F_T.$$

If, in addition,

$$\sum_{i=1}^{n} c_i \leq U_C$$

throughout the operational range, we say the witness is *range-bounded* by U_C.

Remark F.22 (Interpretation of the contamination terms). The parameter $\overline{\Delta}_A$ measures the worst upward drift of the nominally unaffected class, while $\overline{\Lambda}_A$ measures the total spillover mass absorbed by that class. In a simulator study, both can arise without changing the solver: local queue coupling, tie-breaking side effects, or shared downstream bottlenecks can all raise class-A costs even when the primary provenance error still falls on class B.

Proposition F.23 (Min–max lower envelope under bounded drift contamination)**.** *Under Definition F.21,*

$$\mathrm{MM}_T \geq \frac{\underline{d}}{n_B} F_T - \overline{\Delta}_A.$$

Consequently, on any operational range with

$$F_T \geq F_{\min} > 0 \qquad \text{and} \qquad \frac{\underline{d}}{n_B} > \frac{\overline{\Delta}_A}{F_{\min}},$$

min–max remains proof-carrying with slope

$$\underline{\phi}_{\mathrm{MM}}^{\mathrm{cont}} := \frac{\underline{d}}{n_B} - \frac{\overline{\Delta}_A}{F_{\min}}.$$

Proof. Among the n_B class-B vessels, at least one satisfies

$$\max_{j \in B} c_j \geq c_0 + \frac{1}{n_B} \sum_{j \in B} \delta_j \geq c_0 + \frac{\underline{d}}{n_B} F_T.$$

Every class-A vessel satisfies $c_i \leq c_0 + \overline{\Delta}_A$, so the minimum over all vessels obeys

$$\min_{1 \leq i \leq n} c_i \leq c_0 + \overline{\Delta}_A.$$

Therefore

$$\mathrm{MM}_T = \max_{1 \leq i \leq n} c_i - \min_{1 \leq i \leq n} c_i \geq \frac{\underline{d}}{n_B} F_T - \overline{\Delta}_A.$$

If $F_T \geq F_{\min}$, then

$$\mathrm{MM}_T \geq \left(\frac{\underline{d}}{n_B} - \frac{\overline{\Delta}_A}{F_{\min}} \right) F_T,$$

which is positive exactly under the stated condition. □

Proposition F.24 (Conditional Gini lower envelope under bounded spillover)**.** *Under Definition F.21, if the witness is range-bounded by U_C, then*

$$\mathrm{Gini}_T \geq \frac{\max\{0,\, n_A \underline{d} - n_B \overline{\Lambda}_A\}}{n U_C} F_T.$$

Hence Gini remains proof-carrying whenever the contamination is subcritical *in the sense that*

$$n_A \underline{d} > n_B \overline{\Lambda}_A.$$

Proof. For every ordered pair $(i, j) \in A \times B$, the inequality $|c_j - c_i| \geq c_j - c_i$ gives

$$\sum_{i \in A} \sum_{j \in B} |c_j - c_i| \geq \sum_{i \in A} \sum_{j \in B} (c_j - c_i) = n_A \sum_{j \in B} c_j - n_B \sum_{i \in A} c_i.$$

Using the witness inequalities,

$$n_A \sum_{j \in B} c_j - n_B \sum_{i \in A} c_i \geq n_A \sum_{j \in B} (c_0 + \delta_j) - n_B \sum_{i \in A} (c_0 + \alpha_i) = n_A \sum_{j \in B} \delta_j - n_B \sum_{i \in A} \alpha_i \geq (n_A \underline{d} - n_B \overline{\Lambda}_A) F_T.$$

The same argument applies to the ordered pairs $(j, i) \in B \times A$, so the full Gini numerator satisfies

$$\sum_{i=1}^{n} \sum_{j=1}^{n} |c_i - c_j| \geq 2 \max\{0,\, n_A \underline{d} - n_B \overline{\Lambda}_A\} F_T.$$

By range-boundedness,

$$2n \sum_{i=1}^{n} c_i \leq 2nU_C.$$

Dividing yields the claim. □

Corollary F.25 (Operational-range proof-carrying regime under contamination)**.** *Under Definition F.21, min–max remains proof-carrying on any operational range with $F_T \geq F_{\min} > 0$ and*

$$\underline{\phi}_{\mathrm{MM}}^{\mathrm{cont}} := \frac{\underline{d}}{n_B} - \frac{\overline{\Delta}_A}{F_{\min}} > 0.$$

If the witness is also range-bounded by U_C and

$$\underline{\phi}_{\mathrm{Gini}}^{\mathrm{cont}} := \frac{n_A \underline{d} - n_B \overline{\Lambda}_A}{nU_C} > 0,$$

then Gini remains proof-carrying as well. The pristine class-separated witness of Definition F.18 is the special case $(\overline{\Delta}_A, \overline{\Lambda}_A) = (0, 0)$.

Corollary F.26 (Ratio-normalized contaminated envelope)**.** *Under Definition F.21, suppose there exist nonnegative slope coefficients χ_Δ and χ_Λ such that*

$$\overline{\Delta}_A \leq \chi_\Delta F_T, \qquad \overline{\Lambda}_A \leq \chi_\Lambda F_T.$$

Then

$$\mathrm{MM}_T \geq \left(\frac{\underline{d}}{n_B} - \chi_\Delta \right) F_T.$$

If the witness is also range-bounded by U_C, then

$$\mathrm{Gini}_T \geq \frac{\max\{0,\, n_A \underline{d} - n_B \chi_\Lambda\}}{nU_C} F_T.$$

Hence min–max and, conditionally, Gini remain proof-carrying whenever the corresponding replay-certified slope coefficients are positive.

Proof. Substituting $\overline{\Delta}_A \leq \chi_\Delta F_T$ into Proposition F.23 yields the min–max inequality immediately. Substituting $\overline{\Lambda}_A \leq \chi_\Lambda F_T$ into Proposition F.24 yields the Gini inequality. □

Remark F.27 (Why this relaxation matters for the novelty claim). This is the more demanding fairness test. If the lower envelope survives only on a witness family that requires a cleaner controller, then the paper drifts back toward stacking. By contrast, if the same frozen-solver replay bank yields subcritical contamination coefficients, then the metric-level fairness claim still attaches to the process semantics rather than to a larger architecture.

F.5 Matched-seed replay certification of contamination

The contamination parameters in Definition F.21 are calibrated rather than left as symbolic degrees of freedom. The most conservative calibration keeps the solver frozen and compares two trajectories generated from the same simulator seed: one with provenance-preserving state semantics and one with provenance-erased aggregation. This turns contamination into an observable process-semantic delta rather than an architecture delta.

Definition F.28 (Matched-seed replay contamination record)**.** Fix a simulator seed and a frozen controller family. Let $\mathbf{c}^{\mathrm{obs}}$ and $\mathbf{c}^{\mathrm{agg}}$ be the realized per-vessel service-cost vectors under, respectively, the observer-aware and provenance-erased aggregated state semantics, and let $F_T > 0$ be the paired burden count on the aggregated episode. For vessel classes A and B, define

$$\Delta^{\mathrm{pair}}_{A,\max} := \max_{i\in A}(c_i^{\mathrm{agg}} - c_i^{\mathrm{obs}})_+,$$

$$\Lambda^{\mathrm{pair}}_{A} := \sum_{i\in A}(c_i^{\mathrm{agg}} - c_i^{\mathrm{obs}})_+,$$

and

$$D^{\mathrm{pair}}_{B} := \sum_{j\in B}(c_j^{\mathrm{agg}} - c_j^{\mathrm{obs}})_+.$$

Define the observer-aware cross-class baseline slack by

$$S^{\mathrm{obs}}_{AB} := \left(\max_{i\in A} c_i^{\mathrm{obs}} - \min_{j\in B} c_j^{\mathrm{obs}}\right)_+.$$

The normalized contamination / baseline-slack ratios are

$$Q_\Delta := \frac{\Delta^{\mathrm{pair}}_{A,\max}}{F_T}, \qquad Q_\Lambda := \frac{\Lambda^{\mathrm{pair}}_{A}}{F_T}, \qquad Q_S := \frac{S^{\mathrm{obs}}_{AB}}{F_T},$$

and the normalized paired burden ratio is

$$R_B := \frac{D^{\mathrm{pair}}_{B}}{F_T}.$$

Episodes with $F_T = 0$ are excluded from the ratio calibration and satisfy the later lower-envelope inequalities trivially.

Definition F.29 (Witness-dominating matched-seed replay bank)**.** A matched-seed replay bank is *witness-dominating* on an operational range if every aggregated episode on that range admitting Definition F.21 has witness coefficients $(\alpha_i)_{i\in A}$ and $(\delta_j)_{j\in B}$ for which the matched observer-aware replay under the same seed satisfies

$$\max_{i\in A}\alpha_i \le \Delta^{\mathrm{pair}}_{A,\max}, \qquad \sum_{i\in A}\alpha_i \le \Lambda^{\mathrm{pair}}_{A}, \qquad \sum_{j\in B}\delta_j \ge D^{\mathrm{pair}}_{B}.$$

Thus the paired replay upper-bounds class-A contamination and lower-bounds class-B burden relative to the latent witness coefficients required by the structural theorems.

Remark F.30 (Why the witness-dominating condition is the right caution point). Definition F.29 is intentionally strong. It is the point at which the fairness-side route can fail without collapsing the process-semantics claim into vague controller stacking. If the matched-seed observer-aware replay does not dominate the latent witness baseline on the operational range of interest, then the fairness-side lower envelopes are reported as structural diagnostics rather than proof-carrying observables.

Definition F.31 (Upper-confidence contamination quantiles)**.** Consider only matched-seed replay pairs with $F_T > 0$. Let

$$Q_{\Delta,(1)} \le \cdots \le Q_{\Delta,(M_+)}, \qquad Q_{\Lambda,(1)} \le \cdots \le Q_{\Lambda,(M_+)}, \qquad Q_{S,(1)} \le \cdots \le Q_{S,(M_+)}$$

be the order statistics of the contamination / baseline-slack ratios from Definition F.28. For a target tail level $\beta \in (0,1)$ and confidence level $1-\alpha \in (0,1)$, define

$$\ell_{\alpha,\beta} := \min\left\{ \ell \in \{1,\dots,M_+\} : \sum_{j=\ell}^{M_+} \binom{M_+}{j} (1-\beta)^j \beta^{M_+-j} \le \alpha \right\}.$$

If the set above is empty, upper certification fails at the requested (α,β) pair. Otherwise define

$$\widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta,\alpha} := Q_{\Delta,(\ell_{\alpha,\beta})}, \qquad \widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta,\alpha} := Q_{\Lambda,(\ell_{\alpha,\beta})}, \qquad \widehat{\chi}^{\mathrm{UCB}}_{S,\beta,\alpha} := Q_{S,(\ell_{\alpha,\beta})}.$$

We call these one-sided $(1-\alpha)$-confidence upper bounds for $(1-\beta)$-coverage contamination and baseline-slack slopes.

Proposition F.32 (Replay-bank upper-confidence contamination bounds)**.** *Assume Definition F.31 and treat Q_Δ, Q_Λ, and Q_S as exchangeable draws from a replay distribution $\mathbb{P}_{\mathcal{S}}$. Then*

$$\mathbb{P}\Big[\mathbb{P}_{Q_\Delta \sim \mathbb{P}_{\mathcal{S}}}\Big(Q_\Delta \le \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta,\alpha}\Big) \ge 1-\beta\Big] \ge 1-\alpha,$$

and likewise

$$\mathbb{P}\Big[\mathbb{P}_{Q_\Lambda \sim \mathbb{P}_{\mathcal{S}}}\Big(Q_\Lambda \le \widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta,\alpha}\Big) \ge 1-\beta\Big] \ge 1-\alpha,$$

and

$$\mathbb{P}\Big[\mathbb{P}_{Q_S \sim \mathbb{P}_{\mathcal{S}}}\Big(Q_S \le \widehat{\chi}^{\mathrm{UCB}}_{S,\beta,\alpha}\Big) \ge 1-\beta\Big] \ge 1-\alpha.$$

Proof. We prove the claim for Q_Δ; the Q_Λ and Q_S arguments are identical. Let

$$q^\Delta_{1-\beta} := \inf\{q \in \mathbb{R}_+ : \mathbb{P}_{Q_\Delta \sim \mathbb{P}_{\mathcal{S}}}(Q_\Delta \le q) \ge 1-\beta\}$$

be the $(1-\beta)$-upper quantile of the contamination ratio. By minimality of the quantile,

$$p^-_\Delta := \mathbb{P}(Q_\Delta < q^\Delta_{1-\beta}) \le 1-\beta.$$

The event

$$Q_{\Delta,(\ell_{\alpha,\beta})} < q^\Delta_{1-\beta}$$

implies that at least $\ell_{\alpha,\beta}$ of the sampled ratios fall strictly below $q^\Delta_{1-\beta}$. The count of such samples is distributed as $\mathrm{Binomial}(M_+, p^-_\Delta)$ and is therefore stochastically upper-bounded by $\mathrm{Binomial}(M_+, 1-\beta)$. Hence

$$\mathbb{P}\Big(Q_{\Delta,(\ell_{\alpha,\beta})} < q^\Delta_{1-\beta}\Big) \le \sum_{j=\ell_{\alpha,\beta}}^{M_+} \binom{M_+}{j} (1-\beta)^j \beta^{M_+-j} \le \alpha.$$

Therefore, with probability at least $1-\alpha$, we have

$$\widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta,\alpha} = Q_{\Delta,(\ell_{\alpha,\beta})} \geq q^{\Delta}_{1-\beta}.$$

By the definition of the quantile, this implies

$$\mathbb{P}_{Q_\Delta \sim \mathbb{P}_{\mathcal{S}}}\left(Q_\Delta \leq \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta,\alpha}\right) \geq 1-\beta.$$

□

Corollary F.33 (Joint replay-certified contamination regime under a frozen solver)**.** *Let $\alpha_B + \alpha_\Delta + \alpha_\Lambda \leq \alpha$ and $\beta_B + \beta_\Delta + \beta_\Lambda \leq \beta$. Apply Definition F.15 to the paired burden ratio R_B from Definition F.28 to obtain a lower confidence burden slope*

$$\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B} > 0.$$

Suppose the matched-seed replay bank is witness-dominating on an operational range in the sense of Definition F.29, and suppose the aggregated episodes on that range are range-bounded by U_C. Define

$$\widehat{\phi}^{\mathrm{replay}}_{\mathrm{MM}} := \frac{\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}}{n_B} - \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta},$$

$$\widehat{\phi}^{\mathrm{replay}}_{\mathrm{Gini}} := \frac{n_A \widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B} - n_B \widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda}}{n U_C}.$$

Then with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of matched-seed episodes satisfy

$$\mathrm{MM}_T \geq \widehat{\phi}^{\mathrm{replay}}_{\mathrm{MM}} F_T$$

whenever $\widehat{\phi}^{\mathrm{replay}}_{\mathrm{MM}} > 0$. If also $\widehat{\phi}^{\mathrm{replay}}_{\mathrm{Gini}} > 0$, then the same fraction of episodes satisfy

$$\mathrm{Gini}_T \geq \widehat{\phi}^{\mathrm{replay}}_{\mathrm{Gini}} F_T.$$

Thus the contamination coefficients required by Corollary F.26 can be certified from the frozen-solver replay bank rather than inserted by hand.

Proof. Applying Proposition F.16 to the paired burden ratio R_B yields, with confidence at least $1-\alpha_B$, coverage at least $1-\beta_B$ for the lower bound

$$R_B \geq \widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}.$$

Applying Proposition F.32 yields, with confidence at least $1-\alpha_\Delta$, coverage at least $1-\beta_\Delta$ for

$$Q_\Delta \leq \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta},$$

and, with confidence at least $1-\alpha_\Lambda$, coverage at least $1-\beta_\Lambda$ for

$$Q_\Lambda \leq \widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda}.$$

By a union bound over confidence events, all three statements hold simultaneously with confidence at least $1-\alpha$. By a second union bound over coverage fractions, they hold simultaneously on at least a $1-\beta$ fraction of matched-seed episodes. On that event, witness domination gives

$$\overline{\Delta}_A \leq \Delta^{\mathrm{pair}}_{A,\max} = Q_\Delta F_T \leq \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta} F_T,$$

$$\overline{\Lambda}_A \le \Lambda_A^{\mathrm{pair}} = Q_\Lambda F_T \le \widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda} F_T,$$

and

$$\underline{d} F_T \le \sum_{j\in B} \delta_j \quad \text{with} \quad \underline{d} := \widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}.$$

Corollary F.26 then yields the stated lower envelopes for min–max and, when positive, Gini. □

F.6 A witness-free paired-replay route for min–max

The witness-dominating replay condition above remains useful for Gini and for reusing the contaminated-witness structural theorems. But it is still a latent condition. The cleaner min–max route dispenses with that latent step entirely and uses only directly observable matched-seed quantities.

Proposition F.34 (Observable paired-replay min–max lower bound)**.** *For any matched-seed replay pair with $F_T > 0$,*

$$\mathrm{MM}(\mathbf{c}^{\mathrm{agg}}) \ge \frac{D_B^{\mathrm{pair}}}{n_B} - \Delta^{\mathrm{pair}}_{A,\max} - S^{\mathrm{obs}}_{AB}.$$

Equivalently, if a replay episode satisfies

$$R_B \ge d, \qquad Q_\Delta \le \chi_\Delta, \qquad Q_S \le \chi_S,$$

then

$$\mathrm{MM}(\mathbf{c}^{\mathrm{agg}}) \ge \left(\frac{d}{n_B} - \chi_\Delta - \chi_S\right) F_T.$$

Proof. Let $j^\star \in B$ attain the maximum positive paired burden increment, so that

$$(c^{\mathrm{agg}}_{j^\star} - c^{\mathrm{obs}}_{j^\star})_+ \ge \frac{1}{n_B} \sum_{j\in B} (c^{\mathrm{agg}}_j - c^{\mathrm{obs}}_j)_+ = \frac{D_B^{\mathrm{pair}}}{n_B}.$$

Hence

$$c^{\mathrm{agg}}_{j^\star} \ge c^{\mathrm{obs}}_{j^\star} + \frac{D_B^{\mathrm{pair}}}{n_B}.$$

Also, by definition of $\Delta^{\mathrm{pair}}_{A,\max}$,

$$\max_{i\in A} c^{\mathrm{agg}}_i \le \max_{i\in A} c^{\mathrm{obs}}_i + \Delta^{\mathrm{pair}}_{A,\max}.$$

Therefore

$$\begin{aligned}
\mathrm{MM}(\mathbf{c}^{\mathrm{agg}}) = \max_k c^{\mathrm{agg}}_k - \min_\ell c^{\mathrm{agg}}_\ell &\ge c^{\mathrm{agg}}_{j^\star} - \max_{i\in A} c^{\mathrm{agg}}_i \\
&\ge \frac{D_B^{\mathrm{pair}}}{n_B} - \Delta^{\mathrm{pair}}_{A,\max} - \left(\max_{i\in A} c^{\mathrm{obs}}_i - c^{\mathrm{obs}}_{j^\star}\right).
\end{aligned}$$

Since $c^{\mathrm{obs}}_{j^\star} \ge \min_{j\in B} c^{\mathrm{obs}}_j$,

$$\max_{i\in A} c^{\mathrm{obs}}_i - c^{\mathrm{obs}}_{j^\star} \le \left(\max_{i\in A} c^{\mathrm{obs}}_i - \min_{j\in B} c^{\mathrm{obs}}_j\right)_+ = S^{\mathrm{obs}}_{AB}.$$

This proves the first inequality. Dividing by F_T gives the second. □

Corollary F.35 (Witness-free replay-certified min–max under a frozen solver)**.** *Let $\alpha_B + \alpha_\Delta + \alpha_S \le \alpha$ and $\beta_B + \beta_\Delta + \beta_S \le \beta$. Apply Definition F.15 to the paired burden ratio R_B from Definition F.28 to obtain a lower-confidence burden slope*

$$\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B} > 0.$$

Apply Definition F.31 to obtain replay-certified upper bounds

$$\widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta}, \qquad \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}.$$

Define

$$\widehat{\phi}^{\mathrm{pair}}_{\mathrm{MM}} := \frac{\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}}{n_B} - \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta} - \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}.$$

Then with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of matched-seed episodes satisfy

$$\mathrm{MM}(\mathbf{c}^{\mathrm{agg}}) \ge \widehat{\phi}^{\mathrm{pair}}_{\mathrm{MM}} F_T$$

whenever $\widehat{\phi}^{\mathrm{pair}}_{\mathrm{MM}} > 0$.

Proof. Applying Proposition F.16 to R_B yields, with confidence at least $1-\alpha_B$, coverage at least $1-\beta_B$ for

$$R_B \ge \widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}.$$

Applying Proposition F.32 yields, with confidence at least $1-\alpha_\Delta$, coverage at least $1-\beta_\Delta$ for

$$Q_\Delta \le \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta},$$

and, with confidence at least $1-\alpha_S$, coverage at least $1-\beta_S$ for

$$Q_S \le \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}.$$

By a union bound over confidence events, all three statements hold simultaneously with confidence at least $1-\alpha$. By a second union bound over coverage fractions, they hold simultaneously on at least a $1-\beta$ fraction of matched-seed episodes. Proposition F.34 then yields

$$\mathrm{MM}(\mathbf{c}^{\mathrm{agg}}) \ge \left(\frac{\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}}{n_B} - \widehat{\chi}^{\mathrm{UCB}}_{\Delta,\beta_\Delta,\alpha_\Delta} - \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S} \right) F_T.$$

□

F.7 A more conservative witness-free paired-replay route for Gini

The same paired-replay ingredients can also be pushed into Gini without returning to the latent witness-dominating route. The price is a looser certification path: Gini needs an operational-range bound on the total aggregated service cost, and the observer-aware cross-class slack now enters with a larger combinatorial coefficient.

Proposition F.36 (Observable paired-replay Gini lower bound)**.** *Assume the aggregated replay episode is range-bounded by U_C, i.e.*

$$\sum_{k=1}^{n} c_k^{\mathrm{agg}} \le U_C.$$

Then for any matched-seed replay pair with $F_T > 0$,

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \ge \frac{\max\{0,\, n_A D_B^{\mathrm{pair}} - n_B \Lambda_A^{\mathrm{pair}} - n_A n_B S_{AB}^{\mathrm{obs}}\}}{n U_C}.$$

Equivalently, if a replay episode satisfies

$$R_B \ge d, \qquad Q_\Lambda \le \chi_\Lambda, \qquad Q_S \le \chi_S,$$

then

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \ge \frac{\max\{0,\, n_A d - n_B \chi_\Lambda - n_A n_B \chi_S\}}{n U_C}\, F_T.$$

Proof. Let

$$J^+ := \{j \in B : c_j^{\mathrm{agg}} > c_j^{\mathrm{obs}}\}, \qquad g_j := c_j^{\mathrm{agg}} - c_j^{\mathrm{obs}} \quad (j \in J^+), \qquad h_i := (c_i^{\mathrm{agg}} - c_i^{\mathrm{obs}})_+ \quad (i \in A).$$

Then

$$D_B^{\mathrm{pair}} = \sum_{j \in J^+} g_j.$$

For any $i \in A$ and $j \in J^+$ we have

$$c_i^{\mathrm{agg}} \le c_i^{\mathrm{obs}} + h_i, \qquad c_j^{\mathrm{agg}} = c_j^{\mathrm{obs}} + g_j,$$

so

$$c_j^{\mathrm{agg}} - c_i^{\mathrm{agg}} \ge g_j - h_i - (c_i^{\mathrm{obs}} - c_j^{\mathrm{obs}})_+.$$

Hence

$$(c_j^{\mathrm{agg}} - c_i^{\mathrm{agg}})_+ \ge \Big(g_j - h_i - (c_i^{\mathrm{obs}} - c_j^{\mathrm{obs}})_+\Big)_+.$$

Summing over $A \times J^+$ and using $\sum_r z_r^+ \ge (\sum_r z_r)_+$ yields

$$\sum_{i \in A} \sum_{j \in J^+} (c_j^{\mathrm{agg}} - c_i^{\mathrm{agg}})_+ \ge \left(n_A \sum_{j \in J^+} g_j - |J^+| \sum_{i \in A} h_i - \sum_{i \in A} \sum_{j \in J^+} (c_i^{\mathrm{obs}} - c_j^{\mathrm{obs}})_+ \right)_+.$$

By definition,

$$\sum_{j \in J^+} g_j = D_B^{\mathrm{pair}}, \qquad \sum_{i \in A} h_i = \Lambda_A^{\mathrm{pair}}, \qquad |J^+| \le n_B.$$

Moreover,

$$(c_i^{\mathrm{obs}} - c_j^{\mathrm{obs}})_+ \le S_{AB}^{\mathrm{obs}}$$

for every $i \in A$ and $j \in B$, so

$$\sum_{i \in A} \sum_{j \in J^+} (c_i^{\mathrm{obs}} - c_j^{\mathrm{obs}})_+ \le n_A |J^+| S_{AB}^{\mathrm{obs}} \le n_A n_B S_{AB}^{\mathrm{obs}}.$$

Therefore

$$\sum_{i\in A}\sum_{j\in J^+}(c_j^{\text{agg}}-c_i^{\text{agg}})_+ \ge \max\{0,\, n_A D_B^{\text{pair}} - n_B\Lambda_A^{\text{pair}} - n_A n_B S_{AB}^{\text{obs}}\}.$$

Now

$$\sum_{p=1}^{n}\sum_{q=1}^{n}|c_p^{\text{agg}}-c_q^{\text{agg}}| = 2\sum_{p=1}^{n}\sum_{q=1}^{n}(c_q^{\text{agg}}-c_p^{\text{agg}})_+$$

and the full positive-part sum dominates the restricted cross-class sum over $A \times J^+$. Hence the Gini numerator is at least

$$2\max\{0,\, n_A D_B^{\text{pair}} - n_B\Lambda_A^{\text{pair}} - n_A n_B S_{AB}^{\text{obs}}\}.$$

Using the range bound,

$$2n\sum_{k=1}^{n} c_k^{\text{agg}} \le 2nU_C.$$

Dividing yields the first inequality. The second follows by substituting

$$D_B^{\text{pair}} = R_B F_T, \qquad \Lambda_A^{\text{pair}} = Q_\Lambda F_T, \qquad S_{AB}^{\text{obs}} = Q_S F_T.$$

$\square$

Corollary F.37 (Witness-free replay-certified Gini under a frozen solver)**.** *Assume the aggregated episodes on the operational range of interest are range-bounded by U_C. Let*

$$\alpha_B + \alpha_\Lambda + \alpha_S \le \alpha, \qquad \beta_B + \beta_\Lambda + \beta_S \le \beta.$$

Apply Definition F.15 to the paired burden ratio R_B from Definition F.28 to obtain a lower-confidence burden slope

$$\widehat{d}^{\text{LCB}}_{\beta_B,\alpha_B} > 0.$$

Apply Definition F.31 to obtain replay-certified upper bounds

$$\widehat{\chi}^{\text{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda}, \qquad \widehat{\chi}^{\text{UCB}}_{S,\beta_S,\alpha_S}.$$

Define

$$\widehat{\phi}^{\text{pair}}_{\text{Gini}} := \frac{\max\{0,\, n_A \widehat{d}^{\text{LCB}}_{\beta_B,\alpha_B} - n_B\widehat{\chi}^{\text{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda} - n_A n_B \widehat{\chi}^{\text{UCB}}_{S,\beta_S,\alpha_S}\}}{nU_C}.$$

Then with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of matched-seed episodes satisfy

$$\text{Gini}(\mathbf{c}^{\text{agg}}) \ge \widehat{\phi}^{\text{pair}}_{\text{Gini}} F_T.$$

In particular, Gini admits a witness-free paired-replay certification path whenever the replay-certified coefficient above is positive.

Proof. Applying Proposition F.16 to R_B yields, with confidence at least $1-\alpha_B$, coverage at least $1-\beta_B$ for

$$R_B \ge \widehat{d}^{\text{LCB}}_{\beta_B,\alpha_B}.$$

Applying Proposition F.32 yields, with confidence at least $1-\alpha_\Lambda$, coverage at least $1-\beta_\Lambda$ for

$$Q_\Lambda \le \widehat{\chi}^{\text{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda},$$

and, with confidence at least $1-\alpha_S$, coverage at least $1-\beta_S$ for

$$Q_S \le \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}.$$

By a union bound over confidence events, all three statements hold simultaneously with confidence at least $1-\alpha$. By a second union bound over coverage fractions, they hold simultaneously on at least a $1-\beta$ fraction of matched-seed episodes. Proposition F.36 then yields

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \ge \frac{\max\{0,\, n_A\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B} - n_B\widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda} - n_A n_B\widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}\}}{nU_C}\, F_T.$$

□

Remark F.38 (Why this does not promote Gini to the paper's main proof carrier)*.* Corollary F.37 matters because it removes the latent witness-dominating step for Gini as well. But it is still a weaker route than Corollary F.35. The bound is looser, it depends on an operational-range denominator bound U_C, and the observer-aware cross-class slack enters with an $n_A n_B$ factor. Accordingly, Gini remains secondary unless the replay-certified coefficient $\widehat{\phi}^{\mathrm{pair}}_{\mathrm{Gini}}$ stays stably positive under calibration stress. If that coefficient collapses, Gini is treated as a diagnostic or reporting metric rather than a proof-carrying one.

F.8 A denominator-free companion to Gini

If the only obstruction is Gini's denominator, the cautious response is not to force Gini into the paper's proof-carrying core. The cleaner move is to expose the denominator-free quantity that Gini already contains.

Definition F.39 (Absolute pairwise disparity)**.** For any nonnegative service-cost vector $\mathbf{c} = (c_1,\dots,c_n)$, define

$$\mathrm{APD}(\mathbf{c}) := \frac{1}{2n}\sum_{p=1}^{n}\sum_{q=1}^{n}|c_p - c_q|.$$

Equivalently,

$$\mathrm{APD}(\mathbf{c}) = \mathrm{Gini}(\mathbf{c})\sum_{k=1}^{n} c_k.$$

Thus APD is not a different fairness notion from Gini. It is the denominator-free companion obtained by removing the total-cost normalization.

Proposition F.40 (Observable paired-replay lower bound for absolute pairwise disparity)**.** *For any matched-seed replay pair with $F_T > 0$,*

$$\mathrm{APD}(\mathbf{c}^{\mathrm{agg}}) \ge \frac{\max\{0,\, n_A D^{\mathrm{pair}}_B - n_B\Lambda^{\mathrm{pair}}_A - n_A n_B S^{\mathrm{obs}}_{AB}\}}{n}.$$

Equivalently, if a replay episode satisfies

$$R_B \ge d, \qquad Q_\Lambda \le \chi_\Lambda, \qquad Q_S \le \chi_S,$$

then

$$\mathrm{APD}(\mathbf{c}^{\mathrm{agg}}) \ge \frac{\max\{0,\, n_A d - n_B\chi_\Lambda - n_A n_B\chi_S\}}{n}\, F_T.$$

Proof. The proof is the numerator part of Proposition F.36 before any denominator bound is applied. That proof shows

$$\sum_{p=1}^{n}\sum_{q=1}^{n}|c_p^{\mathrm{agg}}-c_q^{\mathrm{agg}}|\geq 2\max\{0,\, n_A D_B^{\mathrm{pair}}-n_B\Lambda_A^{\mathrm{pair}}-n_A n_B S_{AB}^{\mathrm{obs}}\}.$$

Dividing by $2n$ yields

$$\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})\geq\frac{\max\{0,\, n_A D_B^{\mathrm{pair}}-n_B\Lambda_A^{\mathrm{pair}}-n_A n_B S_{AB}^{\mathrm{obs}}\}}{n}.$$

Substituting

$$D_B^{\mathrm{pair}}=R_B F_T,\qquad \Lambda_A^{\mathrm{pair}}=Q_\Lambda F_T,\qquad S_{AB}^{\mathrm{obs}}=Q_S F_T$$

gives the normalized form. □

Corollary F.41 (Witness-free replay-certified absolute pairwise disparity under a frozen solver)**.** *Let*

$$\alpha_B+\alpha_\Lambda+\alpha_S\leq\alpha,\qquad \beta_B+\beta_\Lambda+\beta_S\leq\beta.$$

Apply Definition F.15 to the paired burden ratio R_B from Definition F.28 to obtain a lower-confidence burden slope

$$\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}>0.$$

Apply Definition F.31 to obtain replay-certified upper bounds

$$\widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda},\qquad \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}.$$

Define

$$\widehat{\phi}^{\mathrm{pair}}_{\mathrm{APD}}:=\frac{\max\{0,\, n_A\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}-n_B\widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda}-n_A n_B\widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}\}}{n}.$$

Then with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of matched-seed episodes satisfy

$$\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})\geq\widehat{\phi}^{\mathrm{pair}}_{\mathrm{APD}}F_T.$$

In particular, APD *admits a witness-free replay certification path with no operational-range denominator cap.*

Proof. Applying Proposition F.16 to R_B yields, with confidence at least $1-\alpha_B$, coverage at least $1-\beta_B$ for

$$R_B\geq\widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B}.$$

Applying Proposition F.32 yields, with confidence at least $1-\alpha_\Lambda$, coverage at least $1-\beta_\Lambda$ for

$$Q_\Lambda\leq\widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda},$$

and, with confidence at least $1-\alpha_S$, coverage at least $1-\beta_S$ for

$$Q_S\leq\widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}.$$

By a union bound over confidence events, all three statements hold simultaneously with confidence at least $1-\alpha$. By a second union bound over coverage fractions, they hold simultaneously on at least a $1-\beta$ fraction of matched-seed episodes. Proposition F.40 then yields

$$\mathrm{APD}(\mathbf{c}^{\mathrm{agg}}) \geq \frac{\max\{0,\, n_A \widehat{d}^{\mathrm{LCB}}_{\beta_B,\alpha_B} - n_B \widehat{\chi}^{\mathrm{UCB}}_{\Lambda,\beta_\Lambda,\alpha_\Lambda} - n_A n_B \widehat{\chi}^{\mathrm{UCB}}_{S,\beta_S,\alpha_S}\}}{n} F_T.$$

□

Remark F.42 (Why APD is a proof-carrying companion, not a new headline metric). Corollary F.41 matters because it removes the operational-range denominator cap from the Gini-adjacent route. But that does not mean APD should replace Gini in reporting. The right interpretation is narrower: APD is the proof-carrying companion of Gini when denominator instability makes the normalized statistic too brittle. If APD certifies while Corollary F.37 fails, the correct conclusion is not that the process-semantic disparity disappeared; it is that the denominator side of Gini is not stable enough to carry proof on that operational range.

Remark F.43 (How the Gini/APD split should be used). The clean fairness hierarchy is now: direct service proxy first, witness-free min–max second, and the Gini/APD pair only after that. Within that pair, Gini remains the more familiar headline statistic, while APD is the safer proof carrier because it depends only on the cross-class numerator structure. This keeps the novelty claim attached to process semantics and replay-certified burden, not to a particular normalization convention.

F.9 Replay-certifying the Gini/APD denominator gap

The remaining uncertainty in the Gini/APD pair is now local. Both statistics share the same cross-class numerator, but only Gini divides by the total aggregated service cost. The technical question is therefore not whether the two notions disagree philosophically. The technical question is whether the denominator side is stable enough on the positive-burden operational range that Gini and APD differ only by an almost-constant replay-certified scale.

Definition F.44 (Positive-burden total-cost ratio)**.** For any matched-seed replay pair with $F_T > 0$, define the aggregated total-cost ratio

$$Q_C := \frac{\sum_{k=1}^{n} c_k^{\mathrm{agg}}}{F_T}.$$

Over a replay bank, let

$$Q_{C,(1)} \leq \cdots \leq Q_{C,(M_+)}$$

be the order statistics of these positive-burden ratios.

Definition F.45 (Replay-certified denominator band)**.** Fix tail/confidence pairs (β_L, α_L) and (β_U, α_U) in $(0,1)^2$. Using the same lower-tail index construction as Definition F.15 and the same upper-tail index construction as Definition F.31, applied now to the order statistics of Q_C, define

$$\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L} := Q_{C,(k_{\alpha_L,\beta_L})}, \qquad \widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U} := Q_{C,(\ell_{\alpha_U,\beta_U})}.$$

If either certification index set is empty, denominator-band certification fails at the requested confidence/coverage levels. Otherwise we say the replay bank certifies a positive-burden denominator band whenever

$$0 < \widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L} \leq \widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U}.$$

Define the associated replay-certified denominator distortion factor and normalized gap by

$$\Gamma_C^{\mathrm{replay}} := \frac{\widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U}}{\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L}}, \qquad \Delta_C^{\mathrm{replay}} := 1 - \frac{1}{\Gamma_C^{\mathrm{replay}}}.$$

Proposition F.46 (Replay-certified denominator band on the positive-burden operational range)**.** *Assume the ratios Q_C from Definition F.44 are exchangeable draws from a replay distribution $\mathbb{P}_{\mathcal{S}}$. Let*

$$\alpha_L + \alpha_U \leq \alpha, \qquad \beta_L + \beta_U \leq \beta.$$

If the denominator-band certification in Definition F.45 succeeds, then with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of positive-burden replay episodes satisfy

$$\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L} \leq Q_C \leq \widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U}.$$

Proof. The lower statement is the same order-statistic argument as Proposition F.16, with R replaced by Q_C. The upper statement is the same order-statistic argument as Proposition F.32, with Q_Δ replaced by Q_C. A union bound over the two confidence events yields confidence at least $1-\alpha$, and a second union bound over the two coverage statements yields coverage at least $1-\beta$. ☐

Proposition F.47 (Replay-certified Gini/APD conversion)**.** *Fix a positive-burden replay episode satisfying the denominator band of Proposition F.46. Then*

$$\frac{\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})}{\widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U} F_T} \leq \mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \leq \frac{\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})}{\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L} F_T}.$$

Equivalently, on every such episode the Gini/APD conversion uncertainty is multiplicatively controlled by $\Gamma_C^{\mathrm{replay}}$:

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \in \left[\frac{1}{\Gamma_C^{\mathrm{replay}}}, 1\right] \frac{\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})}{\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L} F_T}.$$

Proof. By Definition F.39,

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) = \frac{\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})}{\sum_{k=1}^n c_k^{\mathrm{agg}}} = \frac{\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})}{Q_C F_T}.$$

Applying the denominator band

$$\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L} \leq Q_C \leq \widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U}$$

and dividing through yields the first inequality. The second is just the same inequality rewritten in terms of

$$\Gamma_C^{\mathrm{replay}} = \frac{\widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U}}{\widehat{q}^{\mathrm{LCB}}_{C,\beta_L,\alpha_L}}.$$

☐

Corollary F.48 (Replay-certified inheritance of Gini from APD)**.** *Let the confidence/coverage budgets used in Corollary F.41 and Proposition F.46 sum to at most (α, β). If both certifications* *succeed, then with confidence at least $1-\alpha$ over the replay bank, at least a $1-\beta$ fraction of positive-burden matched-seed episodes satisfy*

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \geq \frac{\widehat{\phi}^{\mathrm{pair}}_{\mathrm{APD}}}{\widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U}}.$$

Hence a denominator-free APD certificate can be converted into a Gini certificate without imposing a fixed exogenous range cap; the conversion quality is governed by the replay-certified denominator distortion factor $\Gamma^{\mathrm{replay}}_C$.

Proof. Corollary F.41 yields, on the certified coverage event,

$$\mathrm{APD}(\mathbf{c}^{\mathrm{agg}}) \geq \widehat{\phi}^{\mathrm{pair}}_{\mathrm{APD}} F_T.$$

Proposition F.47 gives

$$\mathrm{Gini}(\mathbf{c}^{\mathrm{agg}}) \geq \frac{\mathrm{APD}(\mathbf{c}^{\mathrm{agg}})}{\widehat{q}^{\mathrm{UCB}}_{C,\beta_U,\alpha_U} F_T}.$$

Combining these inequalities yields the claim. The confidence and coverage levels follow by union bounds over the APD and denominator-band certification events. □

Remark F.49 (How to interpret the replay-certified denominator distortion). The quantity $\Gamma^{\mathrm{replay}}_C$ is the missing operational diagnostic. If $\Gamma^{\mathrm{replay}}_C \approx 1$ on the range of interest, then Gini and APD differ only by an almost-constant replay-certified scale and the choice between them is mostly about reporting convention. If $\Gamma^{\mathrm{replay}}_C$ is wide, then denominator instability is itself a measured twin property: APD may still carry proof while Gini should fall back to reporting or diagnostic status. That is the right place for caution. It attaches the weakness to a certified denominator effect, not to the process-semantic numerator structure.

Remark F.50 (Why this is the cleaner novelty test for min–max). Corollary F.35 is stronger, conceptually, than Corollary F.33 for the min–max statistic. It uses the same frozen controller, the same seed distribution, and only directly observable paired-replay quantities. No latent witness coefficients need to be reconstructed or dominated. If this witness-free route is not certified, min–max is reported as a diagnostic statistic rather than a proof-carrying claim.

Remark F.51 (Why the witness-dominating route is still useful). Corollary F.33 now sits in a more clearly auxiliary role. It is still useful for tying the replay path back to the contaminated-witness structural theorems and for checking whether the stricter structural story agrees with the fully observable paired-replay one. But it is not needed for the cleanest fairness-side test. That role belongs to Corollary F.35 and, on a more conservative path, Corollary F.37. The witness-dominating route should therefore be treated as a structural cross-check rather than as part of the primary fairness evidence.

Remark F.52 (What should carry the proof on the fairness side). The proof-carrying fairness observable need not be the same statistic used for headline fairness reporting. A direct service proxy such as disadvantaged-class ready-to-berth delay or class-conditional excess queue hours remains the safest choice, and Proposition F.16 turns its replay calibration into a one-sided confidence statement. However, the paper is now asymmetric in a more precise way. Min–max has two routes: a structural route through the contaminated-witness theorems and a cleaner witness-free route through Corollary F.35. Gini now has two secondary routes as well: the direct witness-free bound of Corollary F.37, and the APD-to-Gini inheritance route of Corollary F.48. The latter is more informative operationally because it turns denominator stability into the replay-certified distortion factor $\Gamma_C^{\mathrm{replay}}$. The APD companion in Corollary F.41 remains the safer proof carrier because it depends only on the cross-class numerator structure. Outside those explicit conditions, headline disparity statistics are reported as diagnostics rather than proof-carrying observables. This keeps the novelty claim attached to process semantics rather than to a chosen fairness transformer.

Proposition F.53 (Conditional simulator-grounded budget lift)**.** *Fix the two-class berth-admission family of Theorem F.3. Suppose Proposition F.13 holds with twin-native idling constant $\underline{\epsilon}_{\mathcal{S}} > 0$, and suppose the chosen fairness proxy $\mathcal{D}_T$ is simulator-certified at slope $\underline{\phi}_{\mathcal{S}} > 0$ on the operational range realized by that family in the sense of Definition F.14. Then every aggregated policy $\pi \in \Pi_{\mathrm{agg}}$ satisfies*

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}_{\mathcal{S}}}\,\mathcal{E}_T^{\mathrm{idle}}(\pi) + \frac{1}{2\underline{\phi}_{\mathcal{S}}}\,\mathcal{D}_T(\pi) + \frac{1}{2}I_T(\pi) \geq \frac{T}{4}.$$

Consequently, Corollary F.10 applies with $(\underline{\epsilon}, \underline{\phi}) = (\underline{\epsilon}_{\mathcal{S}}, \underline{\phi}_{\mathcal{S}})$ whenever the budgets are evaluated on the same operational range.

Proof. By Proposition F.13,

$$\mathcal{E}_T^{\mathrm{idle}}(\pi) \geq \underline{\epsilon}_{\mathcal{S}} E_T(\pi).$$

By simulator certification of the fairness proxy on the operational range,

$$\mathcal{D}_T(\pi) \geq \underline{\phi}_{\mathcal{S}} F_T(\pi).$$

Substituting these inequalities into Corollary F.5 yields

$$V_T(\pi) + \frac{1}{2\underline{\epsilon}_{\mathcal{S}}}\,\mathcal{E}_T^{\mathrm{idle}}(\pi) + \frac{1}{2\underline{\phi}_{\mathcal{S}}}\,\mathcal{D}_T(\pi) + \frac{1}{2}I_T(\pi) \geq \frac{T}{4}.$$

The final statement follows by replacing $(\underline{\epsilon}, \underline{\phi})$ with $(\underline{\epsilon}_{\mathcal{S}}, \underline{\phi}_{\mathcal{S}})$ in Corollary F.10. □

Remark F.54 (Why this is more careful than a blanket fairness theorem)*.* The simulator-grounded lift is deliberately asymmetric. The emission side is inherited directly from the twin's idling semantics. The fairness side is not asserted universally; it is promoted only after the lower-envelope check succeeds on the relevant operational range. If that envelope fails, the stronger fairness-lift claim is removed and the narrower safety–emission or safety–emission–intervention statement is used.

F.10 Robustifying replay certification against dependence and seed shift

The replay-certified burden, contamination, and denominator statements above all use order statistics under an exchangeability assumption. That is the cleanest starting point, but it is also the first empirical seam a reviewer should question. In a maritime digital twin, matched-seed replays may come in dependent clusters: multiple checkpoints from one long rollout, several weather perturbations attached to the same traffic seed, or batches of episodes produced by one scenario-generator state. A second seam is deployment shift: the seed law used in operations may be near, but not identical to, the calibration law used to build the replay bank. Neither issue should be hidden behind architecture gains. The conservative response is to move the certification protocol itself to the block level and then state explicitly how much coverage is lost under bounded seed shift.

Definition F.55 (Seed blocks and conservative block summaries)**.** Fix a replay bank generated under one frozen solver and partition it into disjoint *seed blocks*

$$\mathcal{B}_1, \dots, \mathcal{B}_M,$$

where each block may contain multiple matched-seed episodes produced from one seed family, one contiguous rollout segment, or one scenario-generator state. Let $Z(e)$ be any episode-level statistic used in a replay certificate.

For a statistic that should appear in a lower certificate, define the conservative block summary

$$\underline{Z}_m := \inf_{e \in \mathcal{B}_m} Z(e).$$

For a statistic that should appear in an upper certificate, define the conservative block summary

$$\overline{Z}_m := \sup_{e \in \mathcal{B}_m} Z(e).$$

Thus every lower-certificate quantity is pessimistically reduced inside each block, and every upper-certificate quantity is pessimistically inflated inside each block.

Definition F.56 (Blockwise replay certification)**.** For any lower-certified statistic Z, apply the order-statistic construction of Definition F.15 to the block summaries $\underline{Z}_1, \dots, \underline{Z}_M$, and denote the resulting blockwise lower-confidence bound by

$$\widehat{z}^{\text{blk-LCB}}_{\beta,\alpha}.$$

For any upper-certified statistic U, apply the order-statistic construction of Definition F.31 to the block summaries $\overline{U}_1, \dots, \overline{U}_M$, and denote the resulting blockwise upper-confidence bound by

$$\widehat{u}^{\text{blk-UCB}}_{\beta,\alpha}.$$

For denominator certification, apply Definition F.45 to the block summaries of Q_C and write the resulting band and distortion factor as

$$\widehat{q}^{\text{blk-LCB}}_{C,\beta_L,\alpha_L}, \qquad \widehat{q}^{\text{blk-UCB}}_{C,\beta_U,\alpha_U}, \qquad \Gamma^{\text{blk}}_C.$$

If the relevant order-statistic indices do not exist at the block count M, then blockwise certification fails at the requested confidence/coverage levels.

Proposition F.57 (Within-block dependence robustification)**.** *Assume the seed blocks from Definition F.55 are exchangeable draws from a block law $\mathbb{P}^{\text{blk}}_{\mathcal{S}}$, while episodes within each block may be arbitrarily dependent. Then every order-statistic certification result used in this section remains valid after replacing episode-level statistics by their conservative block summaries. In particular, with confidence at least $1-\alpha$ over the block bank, at least a $1-\beta$ fraction of seed blocks satisfy the corresponding certified lower/upper bounds.*

Moreover, on every certified block, every episode in that block inherits the same bound.

Proof. The previous proofs use only exchangeability of the objects fed to the order statistics. Replacing episode-level observations by the conservative block summaries from Definition F.55 therefore yields the same order-statistic arguments at the block level. The final inheritance statement is immediate: if $\underline{Z}_m \ge c$, then every episode $e \in \mathcal{B}_m$ satisfies $Z(e) \ge c$; if $\overline{U}_m \le c$, then every episode $e \in \mathcal{B}_m$ satisfies $U(e) \le c$. □

Definition F.58 (Bounded seed-shift radius)**.** Let $\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}$ denote a deployment block law on the same block-summary space used for calibration. We say deployment is $\varepsilon_{\mathrm{seed}}$-*close* to calibration whenever

$$d_{\mathrm{TV}}\Big(\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}, \mathbb{P}^{\mathrm{blk}}_{\mathcal{S}}\Big) \le \varepsilon_{\mathrm{seed}},$$

where d_{TV} is total-variation distance.

Proposition F.59 (Seed-shift transfer of certified block events)**.** *Let E be any measurable event on the block-summary space. If*

$$\mathbb{P}^{\mathrm{blk}}_{\mathcal{S}}(E) \ge 1 - \beta$$

and Definition F.58 holds with radius $\varepsilon_{\mathrm{seed}}$, then

$$\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}(E) \ge 1 - \beta - \varepsilon_{\mathrm{seed}}.$$

Thus any replay-certified block event transfers to deployment at the cost of an explicit coverage degradation of at most $\varepsilon_{\mathrm{seed}}$.

Proof. By the defining property of total-variation distance,

$$\big|\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}(E) - \mathbb{P}^{\mathrm{blk}}_{\mathcal{S}}(E)\big| \le d_{\mathrm{TV}}\Big(\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}, \mathbb{P}^{\mathrm{blk}}_{\mathcal{S}}\Big) \le \varepsilon_{\mathrm{seed}}.$$

Hence

$$\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}(E) \ge \mathbb{P}^{\mathrm{blk}}_{\mathcal{S}}(E) - \varepsilon_{\mathrm{seed}} \ge 1 - \beta - \varepsilon_{\mathrm{seed}}.$$

□

Corollary F.60 (Shift-aware replay certification of fairness and denominator events)**.** *Fix any replay-certified event E built from the direct service proxy, the witness-free min–max route, the witness-free Gini route, the APD route, or the denominator band / Gini–APD conversion route, and suppose its calibration proof has been upgraded to blockwise certification by Proposition F.57. If the resulting block-level coverage is at least $1-\beta$ with confidence at least $1-\alpha$, and deployment is $\varepsilon_{\mathrm{seed}}$-close in the sense of Definition F.58, then with confidence at least $1-\alpha$ over the calibration bank,*

$$\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}(E) \ge 1 - \beta - \varepsilon_{\mathrm{seed}}.$$

In particular, if the denominator-band event is certified blockwise, then the same shift-adjusted coverage statement holds for the blockwise distortion factor $\Gamma^{\mathrm{blk}}_{C}$ and the induced APD-to-Gini conversion interval.

Proof. Apply Proposition F.57 to obtain the block-level calibration statement and then Proposition F.59 with that certified event E. □

Remark F.61 (What this robustness layer does and does not buy)*.* This subsection does not solve arbitrary simulator drift. It does something narrower and more useful: it isolates the empirical fragility of replay certification from the semantic novelty claim. If certification survives only after collapsing to large blocks or if the best plausible shift radius $\varepsilon_{\mathrm{seed}}$ is large, then the direct proxy, min–max, Gini, and APD routes should all weaken accordingly, potentially falling back to reporting-only status. That is acceptable. The point is to attach the weakness to the replay protocol and seed law, not to blur it into a controller-stack story.

F.11 Twin-specific block units and seed-shift diagnostics

The previous subsection made replay certification robust to arbitrary within-block dependence, but it left two practical choices abstract: what counts as a seed block in the maritime twin, and how the seed-shift radius $\varepsilon_{\text{seed}}$ should be estimated. This subsection makes both choices explicit enough to be implemented while keeping the claim conservative. The goal is not to prove that the simulator perfectly represents deployment. The goal is to make the replay-transfer loss a measured, falsifiable property of the twin interface.

Definition F.62 (Maritime replay descriptor and block key)**.** For each matched-seed replay episode e, let

$$D(e) = (T_e, W_e, \Delta_e, H_e, M_e, V_e)$$

be its *maritime replay descriptor*, where:

- T_e is a traffic-family descriptor, such as vessel-arrival pattern, cargo-class mix, and demand intensity bucket;
- W_e is a weather / metocean regime descriptor, such as wind, tide, visibility, and storm bucket;
- Δ_e is a disruption-family descriptor, such as crane outage, berth closure, customs delay, ETA shock, or yard congestion bucket;
- H_e is the decision-horizon and epoch-granularity descriptor;
- M_e is the terminal / operating-mode descriptor, such as berth cluster, equipment policy, or human-dispatch mode;
- V_e is the scenario-generator and simulator-version identifier.

A *maritime block key* is a pre-registered coarsening map

$$K : \mathcal{D} \to \mathcal{K},$$

where $\mathcal{K}$ is a finite set of block labels. The seed blocks used in Definition F.55 are then

$$\mathcal{B}_k := \{e : K(D(e)) = k\}, \qquad k \in \mathcal{K}.$$

The map K must be fixed before computing any replay certificate.

Remark F.63 (Why the block key is part of the process claim). The block key in Definition F.62 is not a learned representation and does not modify the solver. It is a measurement convention for the replay bank. That distinction is important: if certificates only hold after changing the controller, the evidence is architectural; if they hold under a frozen controller after changing only the scenario semantics and block summaries, the evidence is process-semantic.

Definition F.64 (Scenario-key and within-key seed shift)**.** Let $\mathbb{P}_{\text{cal}}$ and $\mathbb{P}_{\text{dep}}$ be calibration and deployment laws on the block-summary space, and let K be the maritime block key. Write p_{cal} and p_{dep} for the induced laws on $\mathcal{K}$, and write $\mathbb{P}_{\text{cal}}(\cdot \mid k)$ and $\mathbb{P}_{\text{dep}}(\cdot \mid k)$ for the corresponding conditional block-summary laws. Define

$$\varepsilon_{\text{key}} := d_{\text{TV}}(p_{\text{cal}}, p_{\text{dep}}),$$

Key component	Example buckets	Reason for blocking
Traffic family T	Low / medium / high arrival intensity; cargo-class mix; feeder vs mainline profile	Controls queue pressure and the class-conditional service-cost baseline.
Weather regime W	Calm, visibility-constrained, wind/tide-constrained, storm-disrupted	Controls idling, berth feasibility, and the value of verification.
Disruption family Δ	Crane outage, berth closure, ETA conflict, customs delay, yard congestion	Controls which observer–situation contradictions are operationally critical.
Horizon H	Episode length and decision epoch	Prevents short-horizon certificates from being reused as long-horizon guarantees.
Mode M	Terminal cluster, dispatch policy, authority mode	Captures changes in who can certify, escalate, or override.
Version V	Simulator build, scenario-generator hash, data snapshot	Prevents silent mixture of non-comparable replay regimes.

Table 11: A concrete maritime block key for replay certification. The key is intentionally operational rather than architectural: it groups episodes by scenario-generation and port-operating conditions, not by controller internals.

and define a residual within-key shift budget $\varepsilon_{\text{within}}$ such that

$$\sum_{k\in\mathcal{K}} \min\{p_{\text{cal}}(k), p_{\text{dep}}(k)\} d_{\text{TV}}(\mathbb{P}_{\text{cal}}(\cdot \mid k), \mathbb{P}_{\text{dep}}(\cdot \mid k)) \le \varepsilon_{\text{within}}.$$

The twin-specific seed-shift radius is

$$\varepsilon_{\text{seed}}^{\text{twin}} := \min\{1, \varepsilon_{\text{key}} + \varepsilon_{\text{within}}\}.$$

Proposition F.65 (Key/residual decomposition of seed shift)**.** *Under Definition F.64,*

$$d_{\text{TV}}(\mathbb{P}_{\text{cal}}, \mathbb{P}_{\text{dep}}) \le \varepsilon_{\text{seed}}^{\text{twin}}.$$

Consequently, Proposition F.59 can be applied with $\varepsilon_{\text{seed}} = \varepsilon_{\text{seed}}^{\text{twin}}$.

Proof. Use the standard coupling decomposition. First couple the block keys drawn from p_{cal} and p_{dep} maximally; the mismatch probability is $d_{\text{TV}}(p_{\text{cal}}, p_{\text{dep}}) = \varepsilon_{\text{key}}$. Conditional on matching key k, couple the block summaries maximally under the two conditional laws; the conditional mismatch probability is the corresponding conditional total-variation distance. Averaging over the matched-key mass gives at most $\varepsilon_{\text{within}}$. Hence the total mismatch probability is at most $\varepsilon_{\text{key}} + \varepsilon_{\text{within}}$, which upper-bounds total variation between the full block-summary laws. Capping at one gives the stated radius. □

Definition F.66 (Finite-key upper-confidence radius)**.** Suppose $\mathcal{K}$ has cardinality K_0. Let n_{cal} calibration blocks and n_{dep} deployment-comparison blocks be summarized by empirical key laws $\widehat{p}_{\text{cal}}$ and $\widehat{p}_{\text{dep}}$. For confidence level $1-\alpha$, define

$$\rho(n, K_0, \alpha) := \frac{K_0}{2}\sqrt{\frac{\log(4K_0/\alpha)}{2n}},$$

and

$$\widehat{\varepsilon}_{\text{key}}^{\text{UCB}} := \min\{1, d_{\text{TV}}(\widehat{p}_{\text{cal}}, \widehat{p}_{\text{dep}}) + \rho(n_{\text{cal}}, K_0, \alpha) + \rho(n_{\text{dep}}, K_0, \alpha)\}\,.$$

Given an independently justified residual bound $\widehat{\varepsilon}^{\mathrm{UCB}}_{\mathrm{within}}$, define

$$\widehat{\varepsilon}^{\mathrm{twin,UCB}}_{\mathrm{seed}} := \min\{1, \widehat{\varepsilon}^{\mathrm{UCB}}_{\mathrm{key}} + \widehat{\varepsilon}^{\mathrm{UCB}}_{\mathrm{within}}\}.$$

Proposition F.67 (Conservative finite-key seed-shift certificate)**.** *Assume the calibration and deployment-comparison block keys are independent samples from their respective finite key laws. Then, with probability at least $1-\alpha$ over the key samples,*

$$\varepsilon_{\mathrm{key}} \le \widehat{\varepsilon}^{\mathrm{UCB}}_{\mathrm{key}}.$$

If the residual bound $\widehat{\varepsilon}^{\mathrm{UCB}}_{\mathrm{within}}$ also holds, then

$$d_{\mathrm{TV}}(\mathbb{P}_{\mathrm{cal}}, \mathbb{P}_{\mathrm{dep}}) \le \widehat{\varepsilon}^{\mathrm{twin,UCB}}_{\mathrm{seed}}.$$

Proof. For each key $k \in \mathcal{K}$, Hoeffding's inequality and a union bound imply that, with probability at least $1-\alpha/2$, all calibration key frequencies satisfy

$$|p_{\mathrm{cal}}(k) - \widehat{p}_{\mathrm{cal}}(k)| \le \sqrt{\frac{\log(4K_0/\alpha)}{2n_{\mathrm{cal}}}}.$$

The same bound holds for the deployment-comparison key frequencies with probability at least $1-\alpha/2$. On the intersection of these events, the triangle inequality for total variation gives

$$d_{\mathrm{TV}}(p_{\mathrm{cal}}, p_{\mathrm{dep}}) \le d_{\mathrm{TV}}(\widehat{p}_{\mathrm{cal}}, \widehat{p}_{\mathrm{dep}}) + \frac{1}{2}\sum_{k\in\mathcal{K}} |p_{\mathrm{cal}}(k) - \widehat{p}_{\mathrm{cal}}(k)| + \frac{1}{2}\sum_{k\in\mathcal{K}} |p_{\mathrm{dep}}(k) - \widehat{p}_{\mathrm{dep}}(k)|,$$

which yields the stated upper-confidence radius. The final statement follows from Proposition F.65. □

Definition F.68 (Operational residual diagnostics)**.** The residual term $\widehat{\varepsilon}^{\mathrm{UCB}}_{\mathrm{within}}$ is treated as a stress parameter unless it is supported by within-key diagnostics. For each key k, compare calibration and deployment-comparison blocks using the same frozen solver and the same block-summary coordinates that enter the replay certificates: burden LCBs, contamination UCBs, APD/Gini denominator ratios, intervention counts, and false-certification indicators. If a within-key diagnostic fails, the corresponding key is either split into a finer key or assigned a larger residual budget.

Corollary F.69 (Twin-specific shift-aware replay transfer)**.** *Let E be any blockwise replay-certified event from Corollary F.60. If the maritime block key is fixed as in Definition F.62 and the finite-key seed-shift certificate in Proposition F.67 succeeds, then with confidence reduced by the key-certification level,*

$$\mathbb{P}^{\mathrm{blk}}_{\mathrm{dep}}(E) \ge 1 - \beta - \widehat{\varepsilon}^{\mathrm{twin,UCB}}_{\mathrm{seed}}.$$

Thus the coverage loss used in the replay transfer can be reported as a twin-specific, block-key-dependent quantity rather than as an abstract free parameter.

Remark F.70 (How to use the radius in the paper)*.* The certificate is intentionally one-sided and conservative. A small value of $\widehat{\varepsilon}^{\mathrm{twin,UCB}}_{\mathrm{seed}}$ supports carrying replay-certified fairness or denominator claims to the deployment comparison range. A large value does not refute the process semantics; it says that the chosen calibration bank is not close enough to the deployment seed law to carry those empirical certificates. In that case, the correct response is to expand or reweight the replay bank, split the block key, or weaken the fairness/denominator claims. It is not to add another controller layer.

F.12 Certified actions and explanation soundness

Theorem F.71 (Certified-action soundness)**.** *Fix time t, agent i, and physical action a_i^p. Suppose $S \subseteq C_t$ is a support certificate for a_i^p at decision node $\lambda_i(x_t)$. Then there exists a classical propositional interpretation I such that*

$$I \models \mathrm{Th}_{\lambda_i(x_t)}(S),$$

and in particular

$$I \models Pre_i(a_i^p).$$

Therefore, every certified action admits a semantically sound explanation witness at its decision node.

Proof. Because S is a support certificate, $\mathrm{Th}_{\lambda_i(x_t)}(S)$ is propositionally satisfiable by Definition 6.10. By OSL semantic soundness, satisfiability of the supported theory at a lattice element implies the existence of a classical interpretation satisfying that theory [51]. Thus there exists I such that

$$I \models \mathrm{Th}_{\lambda_i(x_t)}(S).$$

Again by Definition 6.10,

$$Pre_i(a_i^p) \subseteq \mathrm{Th}_{\lambda_i(x_t)}(S),$$

so the same I satisfies every precondition of a_i^p. Hence the emitted certificate is a sound explanation witness for the physical action. □

F.13 Bounded-overhead inheritance

Proposition F.72 (Regret and violation transfer under bounded epistemic interventions)**.** *Let π^{base} be any baseline feasible controller on an* OS-CPOSG *subcase that ignores epistemic exceptions. Let π^{os} be the contradiction-aware wrapper that follows π^{base} except on at most M_T decision epochs, where it replaces the proposed physical action by an epistemic exception act.*

Assume that each such intervention changes cumulative operational reward by at most Δ and changes cumulative constraint cost g_j by at most β_j. Then

$$\mathrm{Reg}_T(\pi^{os}) \leq \mathrm{Reg}_T(\pi^{base}) + \Delta M_T,$$

and for each constraint index j,

$$\mathrm{Viol}_j^{(T)}(\pi^{os}) \leq \mathrm{Viol}_j^{(T)}(\pi^{base}) + \beta_j M_T.$$

Consequently, if $M_T = o(T)$, the wrapper preserves the baseline asymptotic average-regret and average-violation rates.

Proof. Write $R_T(\pi)$ for cumulative reward of policy π. Since π^{os} differs from π^{base} on at most M_T epochs and each deviation changes total reward by at most Δ,

$$R_T(\pi^{os}) \geq R_T(\pi^{base}) - \Delta M_T.$$

Let $R_T^\star$ be the reward of an optimal feasible policy. Then

$$\mathrm{Reg}_T(\pi^{os}) = R_T^\star - R_T(\pi^{os}) \le R_T^\star - R_T(\pi^{base}) + \Delta M_T = \mathrm{Reg}_T(\pi^{base}) + \Delta M_T.$$

The same decomposition applies coordinate-wise to cumulative constraint costs. If the intervention changes the total cost in coordinate j by at most β_j on each of at most M_T epochs, then

$$\mathrm{Viol}_j^{(T)}(\pi^{os}) \le \mathrm{Viol}_j^{(T)}(\pi^{base}) + \beta_j M_T.$$

Dividing by T yields the asymptotic statement whenever $M_T = o(T)$. □

F.14 Contradiction burden and half-life

Definition F.73 (Contradiction burden)**.** The contradiction burden at time t is

$$B_t := \sum_{K \in \mathrm{MCC}(C_t)} \omega(K).$$

This is the total weighted mass of active contradiction components in the current claim base.

Definition F.74 (Clearing time of a contradiction component)**.** Let K be a contradiction component that first appears at time $b(K)$. Its *clearing time* is

$$\begin{aligned} t_{\mathrm{clr}}(K) := \inf \Big\{ t \ge b(K) : K \notin \mathrm{MCC}(C_t) \\ \text{or } \forall i, \forall a_i^p \in A_i^{\mathrm{crit}},\ K \cap \mathsf{Rel}_i(x_t, C_t, a_i^p) \neq \varnothing \\ \Rightarrow a_i^e \in \mathsf{Exc}_i(a_i^p) \Big\}. \end{aligned}$$

Thus a contradiction is cleared once it is either resolved from the claim base or safely routed away from critical physical execution by epistemic exception acts.

Definition F.75 (Shock half-life)**.** For a contradiction shock injected at time t_0, define the shock half-life by

$$\tau_{1/2}(t_0) := \inf \left\{ h \ge 0 : B_{t_0+h} \le \tfrac{1}{2} B_{t_0} \right\}.$$

In episodic evaluation, the reported contradiction half-life is the mean or median of $\tau_{1/2}(t_0)$ over all tagged contradiction shocks.

Remark F.76. If new unrelated contradictions can enter after t_0, then empirical half-life should be computed over the tagged contradiction burden generated by the original shock rather than over all contradictions globally. This avoids penalizing a policy for independent events that arrive after the measured shock.

G Additional related-work comparisons

This section makes the novelty boundary explicit. The question is not whether there exist prior models that reason about other agents, beliefs, common knowledge, or norms. There are several. The question is whether there exists a simulator-coupled constrained multi-agent process in which observer–situation-indexed contradictions are first-class state variables, action preconditions, and audit objects. The closest neighboring lines are the following.

G.1 Interactive belief-state models

I-POMDPs extend POMDPs to multi-agent settings by incorporating models of other agents into the state space and maintaining beliefs over both the physical environment and the models of other agents [28]. That is structurally important because it shows how multi-agent uncertainty can be lifted into the state itself rather than handled as an afterthought. However, the object being maintained is still a nested belief state. Contradictions are not preserved as named components with provenance and lattice coordinates, and action admissibility is not defined in terms of contradiction-aware certification.

G.2 Common-knowledge coordination

MACKRL shows that common knowledge can support powerful decentralized coordination without learned communication protocols, via a hierarchical policy tree conditioned on what groups of agents can independently reconstruct from their observations [19]. That line is close in spirit because it elevates an epistemic notion from background intuition to control primitive. The difference is that MACKRL is built around *agreed* information. OS-CMASP is built around *disagreed* information: provenance-bearing claims that may remain incompatible until the policy spends effort to resolve them, route around them, or escalate them.

G.3 Multi-agent epistemic planning

Epistemic planning work, including planners with common knowledge such as MEPC, studies actions with epistemic preconditions and effects and can explicitly represent higher-order and common-knowledge reasoning [26]. This is the closest line on the knowledge-representation side. The gap is that epistemic planning is usually not framed as stochastic, simulator-coupled, constrained control with cumulative operational objectives, fairness penalties, and online feasibility gating. In other words, epistemic planning is close on semantic content but not on process class.

G.4 Simulator-coupled, norm-governed processes

R-CMASP is the nearest formal neighbor. It augments stochastic games and Dec-POMDPs with simulator-coupled transitions, role-specialized agents, structured observability and communication, and a normative feasibility layer over joint actions [43]. Relative to that model, the proposed delta is sharp:

1. the state contains an explicit claim base over observer–situation lattice nodes;
2. contradiction components are first-class objects obtained by MCC rather than implicit inconsistencies inside local beliefs;
3. admissibility of critical actions depends on support certificates or epistemic exception acts; and
4. the formalism admits an observer-collapse separation theorem showing that erasing provenance can force linear regret.

This is the strongest comparison in the paper. If those four deltas cannot be maintained, the OS-CMASP active regime should be narrowed and the missing delta should be stated explicitly, rather than recasting the work as an OSL module for an existing controller.

G.5 Current CH-MARL / EcoFair baseline

EcoFair-CH-MARL and the broader CHMARL project already provide the domain baseline and deployment narrative for maritime coordination: a two-tier planner / executor split, real-time budgeted optimization, fairness-aware rewards, and an emphasis on explainability, auditability, and human authority [13, 14]. The present work does not attempt to improve that controller within the same ontology. Instead, it asks whether the maritime environment itself should be represented differently: not as a single operational state with noisy observations, but as an operational state plus a contradiction-bearing claim state over observer–situation space.

G.6 Concise novelty statement

The cleanest novelty statement I know how to defend is the following:

> **Defensible novelty statement.** Relative to EcoFair-CH-MARL, the contribution is not a better two-tier controller. Relative to R-CMASP, the contribution is not merely simulator coupling plus norms. Relative to belief-state and common-knowledge models, the contribution is not merely richer epistemic structure. The contribution is a candidate process class in which fragmented truth is explicit in the state, contradiction repair is explicit in the action space, and certification / escalation is explicit in the feasibility relation.

That statement is stronger than "we add OSL to CH-MARL" but narrower than "nothing like this exists"; it is intentionally written as a candidate rather than a settled class claim. The separation theorem is what determines whether the stronger interpretation is deserved.

H Detailed minimal separation benchmark: Berth-1-Conflict

The empirical program begins with a deliberately small benchmark rather than the full 16-port / 50-vessel digital twin. The benchmark instantiates Theorem 7.3 while remaining legible to reviewers. We call it Berth-1-Conflict.

H.1 Benchmark purpose

The benchmark should answer one narrow question:

> When a high-impact maritime action depends on conflicting reports from differently prioritized observers, does preserving observer–situation provenance reduce regret and unsafe actions relative to controllers that only see aggregated content?

Component	Initial benchmark instantiation
Physical state	Berth readiness, crane health, vessel arrival window, queue cost, emissions cost.
Observers	AIS feed, terminal operating system / strategic berth plan, harbor master, crane maintenance system, human dispatcher.
Situations	Strategic planning, local operational update, exception / override context.
Physical acts	Admit vessel, hold vessel, reroute vessel (optional in first version).
Epistemic acts	Query maintenance, request local confirmation, certify action, defer, escalate.
Constraints	Safety violation cost, waiting-time cost, idling-emissions cost, optional fairness term if multiple vessel classes are added later.
Explanation object	Support certificate for any admitted or rerouted action.

Table 12: Proposed first benchmark for instantiating the observer-collapse separation result in a maritime setting.

If the answer is not clearly yes on this minimal benchmark, there is little value in scaling to the full maritime twin.

H.2 Environment skeleton

The environment contains a single arriving vessel and a single berth. The hidden physical state includes berth readiness, crane availability, and vessel ETA consistency. Observers include an AIS feed, a terminal operating system or strategic berth plan, a harbor master report, a crane maintenance system, and a human dispatcher. Situations distinguish at least "global planning", "local operational update", and "override / exception" contexts.

Physical actions are

$$A^p = \{\mathsf{admit}, \mathsf{hold}, \mathsf{reroute}\}.$$

Epistemic actions are

$$A^e = \{\mathsf{queryMaint}, \mathsf{queryLocal}, \mathsf{certify}, \mathsf{defer}, \mathsf{escalate}\}.$$

Critical actions are admit and, when modeled, $\mathsf{reroute}$.

The vocabulary may begin with

$$\Phi = \{ready, \neg ready, craneOK, \neg craneOK, etaOnTime, \neg etaOnTime\}.$$

Each observation is translated into one or more claim records of the form $(\varphi, \ell, w, \tau, \rho)$ and inserted into the claim base before RBP and MCC are applied.

H.3 Controller-visible context map for Berth-1-Conflict

The separation theorem is meaningful only if the hidden operational truth is not leaked through ordinary physical features. We therefore instantiate the visible-context map explicitly.

Let the one-berth latent physical state be

$$x = (b, c, e, q, z, v, m, t),$$

where $b \in \{0, 1\}$ is true berth readiness, $c \in \{0, 1\}$ is true crane availability, e is the realized arrival-window status, q is queue length, z is public weather / disruption regime, v is vessel class, m is operating mode, and t is a time bucket. The controller-visible non-claim context is

$$\chi_{B1}(x) = (q, z, v, m, t, \mathrm{slot}(x), \mathrm{etaBin}(x)),$$

where $\mathrm{slot}(x)$ is the scheduled berth window and $\mathrm{etaBin}(x)$ is the publicly exposed ETA bucket. Crucially, b and c are not components of χ_{B1}; they are available only through claim records generated by the relevant observers.

A minimal authority-conflict pair for the theorem is

$$x^+ = (b = 1, c = 1, e, q, z, v, m, t), \qquad x^- = (b = 0, c = 1, e, q, z, v, m, t),$$

with the same visible context

$$\chi_{B1}(x^+) = \chi_{B1}(x^-).$$

Let ℓ_H denote the harbor-master local-update node and ℓ_P the strategic-plan node, with $\pi_L(\ell_H) > \pi_L(\ell_P)$. Define

$$C^+ = \{(\mathit{ready}, \ell_H, 1, \tau, \rho_H),\ (\neg\mathit{ready}, \ell_P, 1, \tau, \rho_P)\},$$
$$C^- = \{(\mathit{ready}, \ell_P, 1, \tau, \rho_P),\ (\neg\mathit{ready}, \ell_H, 1, \tau, \rho_H)\}.$$

Then

$$\mathrm{Agg}(C^+) = \mathrm{Agg}(C^-),$$

but the dominant support certificate for *ready* exists only in C^+. This is the concrete benchmark instantiation of Theorem 7.3: the aggregated controller sees the same non-claim context and the same unordered content, while the provenance-aware controller sees the authority reversal.

Proposition H.1 (Benchmark realization of the observer-collapse construction)**.** *Under the state pair and claim bases above, any policy in Π_{agg} must assign the same distribution over $\{\mathsf{admit}, \mathsf{hold}\}$ in $s^+ = (x^+, C^+)$ and $s^- = (x^-, C^-)$. A provenance-aware policy can choose* admit *in s^+ and* hold *in s^-. If* admit *requires a dominant support certificate for ready, then* admit *is admissible in s^+ and inadmissible in s^-.*

Proof. The first statement follows from $\chi_{B1}(x^+) = \chi_{B1}(x^-)$ and $\mathrm{Agg}(C^+) = \mathrm{Agg}(C^-)$. The second statement follows because ℓ_H dominates ℓ_P. In C^+, the highest-priority comparable claim supports *ready*, so *ready* has a dominance-safe witness and admit can be certified. In C^-, the highest-priority comparable claim supports $\neg$*ready*, so no dominance-safe witness for *ready* exists. Hence the admissibility status of admit differs across two states that are identical to a label-collapsed policy. □

H.4 Pre-registered maritime dominance order

The dominance relation should be specified before training or replay. Otherwise, the paper risks fitting authority after seeing outcomes. For the first benchmark, use a proposition-family priority table rather than a single universal trust ranking. This is still compatible with Definition 6.4: for each proposition family, instantiate a fixed $\pi_L^{(\varphi)}$ and then use the same lexicographic tie-breakers–priority, recency, credibility–inside that family.

Proposition family	**Dominance order, highest first**	**Reason for pre-registration**
ready	Human dispatcher override > harbor master local update > terminal operating system local berth status > strategic berth plan / schedule-derived readiness	Berth readiness is an operational-local fact; a fresh local authority should dominate a stale global plan.
craneOK	Crane operator or maintenance override > crane maintenance system > terminal operating system equipment status > strategic plan	Crane availability is equipment-local; maintenance provenance should dominate planning assumptions.
etaOnTime	Harbor master / pilot update > fresh AIS-derived ETA > carrier schedule > strategic plan	ETA is trajectory-local and time-sensitive; recency matters after source authority.
weatherSafe	Port safety officer / dispatcher override > local sensor nowcast > port-approved weather feed > strategic weather assumption	Safety-critical weather judgments should prioritize local exception authority and current sensing.

Table 13: Pre-registered proposition-family dominance order for the minimal maritime benchmark. The table is not a learned component; it is part of the environment specification being tested.

For reproducibility, the benchmark should publish the exact rank values, freshness windows, and tie-breaking rules. A minimal version is:

$$\mathrm{rank}_{\varphi}(c) = (\pi_L^{(\varphi)}(\ell), \mathrm{fresh}_{\varphi}(\tau; t), w),$$

ordered lexicographically, where $\mathrm{fresh}_{\varphi}(\tau; t) = 1$ if the claim is inside the pre-registered freshness window for proposition family φ and 0 otherwise. The timestamp τ can still be used as a secondary tie-breaker among fresh claims. This keeps the benchmark from silently converting "authority" into an outcome-fitted score.

Remark H.2 (Typed dominance is a benchmark instantiation, not a new layer)*.* The proposition-family priority table is not an additional controller module. It is an explicit environment-side semantics for how claims become dominance-safe witnesses. If changing this table after seeing results is necessary for separation, the process-class claim should be considered unsupported.

Numerical default for the first pre-registration

The table above gives the qualitative order. For the first implementation, use the following numerical template unless a port-specific operating rule is declared before any training run:

$$\pi_L^{(\varphi)}(\ell) \in \{5, 4, 3, 2, 1\},$$

where 5 denotes authorized human override, 4 denotes the proposition-family local authority, 3 denotes a local system of record, 2 denotes an external operational feed, and 1 denotes a strategic-plan or schedule-derived assumption. Ties are broken by freshness and then credibility.

Family	**Fresh window**	**Stale penalty**	**Default interpretation**
ready	30 minutes	freshness bit drops from 1 to 0	Berth-readiness reports stale quickly enough that a planning assumption should not dominate a fresh local update.
craneOK	15 minutes	freshness bit drops from 1 to 0	Equipment-state claims are treated as short-lived unless maintenance explicitly certifies persistence.
etaOnTime	10 minutes	freshness bit drops from 1 to 0	ETA claims are trajectory-sensitive and should be refreshed frequently under disruption.
$weatherSafe$	10 minutes	freshness bit drops from 1 to 0	Weather/safety claims are short-lived; stale favorable weather must not certify critical entry.

Default freshness windows for the first one-berth benchmark. These are experimental defaults, not claims about a specific port.

The numerical values are intentionally simple. Their role is to make the first benchmark reproducible and falsifiable, not to encode a universal maritime authority model. In the full twin, these values is replaced by documented operational policy or by a sensitivity grid declared before replay.

H.5 Fixed-solver ablation contract

The empirical test must isolate state semantics from architecture. The following contract defines the pre-registered design for the first benchmark.

A1: Same solver. Use the same policy class, optimizer, reward, training budget, random seeds, and action interface across provenance-preserving and provenance-erased conditions.

A2: Same physical observations. Both conditions receive the same $\chi_{B1}(x)$ and the same literal / credibility / timestamp content. The only removed information in the collapsed condition is observer–situation label and provenance.

A3: Same admissibility budget. If the provenance-preserving condition uses certification or escalation for critical admits, the collapsed condition must face the same safety budget. It may hold, query, or escalate, but it cannot execute an uncertified critical admit for free.

A4: Placebo labels. Include a condition in which labels are randomly permuted independently of truth. This should collapse toward the aggregated baseline; otherwise the experiment may be exploiting label artifacts rather than provenance semantics.

A5: Oracle sanity check. Expose b directly through χ in a non-deployment oracle condition; the provenance gap should shrink or disappear.

Condition	Information available	Interpretation
Provenance-preserving	$\chi_{B1}(x)$ plus full claim records $(\varphi, \ell, w, \tau, \rho)$	Tests the proposed process semantics.
Provenance-erased	$\chi_{B1}(x)$ plus $\mathrm{Agg}(C)$	Main collapsed-state baseline.
Labels without gating	Full claim records, but no certificate-based admissibility	Separates provenance-bearing state from feasibility semantics.
Random-label placebo	Full-size labels assigned independently of truth	Tests whether gains rely on meaningful observer–situation structure.
Oracle visible-state	$\chi_{B1}(x)$ augmented with true readiness b	Sanity check; should reduce the provenance advantage.

Ablation conditions for the first benchmark. The solver must remain fixed so that any separation is attributable to process semantics rather than controller growth.

The decision criterion is deliberately strict. The process-semantics claim is empirically supported by the first benchmark only if the provenance-preserving condition separates from provenance-erased and random-label conditions under the same solver, while the oracle condition closes the gap and the benign-agreement scenarios show limited epistemic overhead. If those checks fail, the next analytical step is an active-regime diagnosis: determine whether failure came from leakage through visible context, a non-action-relevant dominance order, insufficient contradiction burden, or a solver that never queries the certificate semantics. The interpretation remains process-semantic; only the benchmark claim narrows.

H.6 Environment-step protocol

To keep the implementation aligned with the theorem, the environment step should be written as a process-semantic transform rather than as a new controller. One episode step is:

E1: Latent update. Maintain latent state $x_t = (b_t, c_t, e_t, q_t, z_t, v_t, m_t, t)$ and expose only $y_t = \chi_{B1}(x_t)$ to the controller.

E2: Claim generation. Generate observer reports using the scenario family. Each report becomes a claim record $(\varphi, \ell, w, \tau, \rho)$.

E3: Epistemic state update. Apply $C_t^+ = \mathrm{RBP}(C_t \cup \Gamma(o_t, a_{t-1}^e))$ and compute $\mathrm{MCC}(C_t^+)$.

E4: Ablation projection. Produce the condition-specific input: full C_t^+, collapsed $\mathrm{Agg}(C_t^+)$, randomly permuted labels, or oracle-augmented χ_{B1}.

E5: Frozen solver call. Use the same solver to propose physical action a_t^p and optional epistemic action a_t^e.

E6: Admissibility check. For critical actions, test dominant support certificates using the pre-registered dominance order. If certification fails, the only admissible alternatives are hold, defer, query, or escalate according to the condition's action interface.

E7: Physical transition and accounting. Apply the accepted action in the simulator and record reward, safety violation, idling-emission proxy, service-delay proxy, epistemic intervention count, contradiction half-life, and certificate validity.

This protocol is the implementation analogue of the non-stacking criterion: only the state projection and feasibility semantics change across ablations; the solver identity does not.

H.7 Minimum viable experiment protocol

A conservative first run should be paired-seed and scenario-balanced.

Protocol item	**Default setting**
Scenario families	Authority conflict, freshness conflict, context conflict, human override, benign agreement.
Pairing	Every random seed is replayed under all ablation conditions.
Training budget	Same number of environment steps, optimizer updates, and random seeds for every learned condition.
Primary endpoint	Zero-violation constrained regret on authority and freshness conflicts.
Secondary endpoints	Critical-action violation rate, false-certification rate, contradiction half-life, intervention cost, waiting time, idling-emission proxy, and service-delay proxy.
Placebo expectation	Random-label placebo should behave like provenance-erased or worse.
Oracle expectation	Oracle visible-state should close most of the provenance gap, verifying that hidden truth was available only through claim provenance.
Benign-agreement expectation	Provenance-preserving certification introduces only limited epistemic overhead when reports agree.

Minimum viable experiment protocol for deciding whether to scale beyond the one-berth benchmark.

The first result to plot should be simple: cumulative zero-violation regret versus horizon for authority-conflict seeds, with one curve per ablation condition. The process-class claim is not strengthened by adding controller complexity; it is strengthened if the same controller separates only when observer–situation provenance is preserved.

H.8 Executable scaffold, paired ablation, and replay adapter

The accompanying materials include a runnable scaffold, `code/berth1_conflict.py`. The file is deliberately dependency-free and implements the theorem-facing constraint that hidden readiness and the other critical precondition truths are excluded from the visible context map $\chi_{B1}(x)$ and can enter the ordinary controller only through claim records. The same dominance-gated solver is used in all non-oracle conditions; the ablation changes only claim-state semantics or certification availability.

The harness uses a paired episode bank: each ablation condition evaluates the same latent trajectory, and randomized placebo operations use separate per-step ablation random streams so they cannot perturb the latent state sequence. The default executable path is preflight rather than evidence generation. Synthetic or replay evaluations must be invoked explicitly after the ablation contract is locked.

The scaffold exposes six conditions:

1. `provenance_preserving`: observer labels, situation labels, ranks, freshness, and provenance are available to the certificate rule;

2. `provenance_erased`: observer–situation labels are erased and the same solver falls back to a safe hold when it cannot certify;
3. `random_label_placebo`: labels are randomly reassigned within proposition families while literals are preserved;
4. `labels_without_gating`: provenance is visible, but certificate-based admissibility is disabled, isolating the safety role of the gate;
5. `oracle_visible_state`: hidden precondition truth is deliberately exposed as a sanity upper bound;
6. `benign_agreement`: claim literals are made coherent while labels are preserved, measuring overhead when contradiction is absent.

Before running anything locally, the paper does not include numerical scaffold outputs. The role of the executable at this stage is to lock the interface and invariants, not to supply evidence. The locked contract is therefore:

C1: No bundled result files. The released package contains source, schema templates, and preflight tooling, but no synthetic summary CSVs or diagnostics JSON that could be mistaken for experiment results.

C2: Preflight by default. Running `berth1_conflict.py` without an explicit `-mode run` performs static checks only: visible-context leakage, replay-schema validity, condition-list consistency, and paired-bank metadata.

C3: Synthetic results require acknowledgement. Running on the built-in synthetic generator requires `-allow-synthetic-results`. This prevents quick scaffolding runs from becoming undeclared evidence.

C4: Replay results require a manifest. Maritime-twin replay should be accompanied by a run manifest recording the replay file hash, condition set, solver version, dominance table version, scenario-key definition, and output prefix.

C5: The same ablation interface must be preserved. The first local run should compare exactly the six conditions listed above; adding a new controller is disallowed until this fixed-solver comparison is reported.

This lock-first posture is part of the non-stacking design. The benchmark is more credible when the ablation interface is fixed before any local result is inspected.

The same script now supports a replay adapter. A maritime twin can emit a claim-level CSV with required columns

`seed,t,ready,scenario,prop,value,observer,situation`,

and optional fields for the other preconditions, visible context keys, credibility, timestamp, and provenance. The command

```
python3 -S berth1_conflict.py -mode run
-replay-csv twin_replay_claims.csv
-out-prefix twin_replay_v1
-manifest twin_replay_v1.manifest.json
```

then runs the same ablation interface on replayed claims. The point is to replace the synthetic generator while preserving the fixed-solver comparison contract.

H.9 Scenario families

The benchmark should contain a small number of interpretable scenario families rather than an undifferentiated random scenario mixture.

S1: Authority conflict. A high-priority local observer says the berth is not ready, while a lower-priority global feed says it is ready.

S2: Freshness conflict. Two observers of similar nominal authority disagree, but one claim is stale.

S3: Context conflict. A proposition holds under the planning situation but not under the current operational situation.

S4: Human override. The automated support set is insufficient or contradictory, forcing escalation to a dispatcher.

S5: Benign agreement. All relevant observers agree, so the epistemic wrapper should stay mostly dormant and preserve baseline efficiency.

These scenarios test not only accuracy under disagreement but also restraint: the system does not over-query or over-escalate when the claim state is already coherent.

H.10 Baselines

The comparison set should be narrow and principled.

Baseline	Purpose
Aggregated-state CH-MARL controller	Uses fused features only; tests whether provenance erasure hurts.
Observer-aware controller without admissibility gating	Sees provenance, but does not require certification or exception acts; isolates the value of state semantics alone.
Full OS-CMASP controller	Uses provenance-bearing claims, contradiction mass, and certification / escalation gating.
R-CMASP-style governed baseline	Adds simulator coupling and norm-gating without OSL-indexed contradiction state; tests the value of the explicit claim-state formulation.

Minimal baseline suite for the first separation-oriented benchmark.

If it is easy to do so, an additional common-knowledge baseline can be included, but only if it is implemented honestly enough to avoid straw-manning MACKRL-style coordination.

H.11 Metrics

The benchmark should report more than reward. The key metrics are:

- cumulative regret relative to the best contradiction-aware policy;
- critical-action violation rate;
- false-certification rate;
- contradiction half-life (time from contradiction appearance to a contradiction-resolved or contradiction-routed state);
- escalation precision (fraction of escalations that are actually needed);
- epistemic intervention cost (queries, deferrals, or escalations per episode);
- throughput / waiting-time / emissions where those are meaningful in the toy setup.

The important visual should be a regret or violation curve showing a clear gap between aggregated-state controllers and provenance-preserving controllers under authority and freshness conflict, while showing only a small overhead gap under benign agreement.

H.12 Hypotheses

The benchmark is strong enough if it tests the following hypotheses.

H1: Separation. Aggregated-state policies incur strictly higher regret than provenance-aware policies under repeated authority conflict.

H2: Safety. Certification / escalation gating reduces unsafe admits relative to observer-aware policies without gating.

H3: Restraint. In benign-agreement scenarios, the full OS-CMASP policy incurs only a small epistemic overhead.

H4: Auditability. The fraction of critical admits accompanied by valid support certificates approaches one whenever escalation is not used.

H.13 Path to the full maritime twin

Only after the minimal benchmark works should the approach be ported into the larger maritime digital twin. The scaling path is straightforward: vessel ETA, berth readiness, crane maintenance, customs release, yard readiness, and weather become claim-generating subsystems; the strategic layer allocates verification and escalation budget in addition to emission and fairness budget; and the operational layer executes physical control while remaining subject to contradiction-aware admissibility. At that point, the evaluation can inherit the original EcoFair metrics and add contradiction-sensitive ones such as contradiction half-life and false-certification rate.

H.14 Experimental lock-in protocol

I Active-regime diagnosis and solver-isolation plan

A strong process-semantics claim requires a clear diagnostic logic. The relevant test is whether the OS-CMASP gain survives when the controller family is held as fixed as possible.

I.1 What would narrow the active OS-CMASP regime?

The following outcomes do not reclassify the paper as a controller add-on. They narrow the active regime and identify which semantic condition must be strengthened.

F1: No state separation. A controller given only $\mathrm{Agg}(C_t)$ matches the observer-aware controller on the separation benchmark without extra violations.

F2: No feasibility separation. Removing certificate-based admissibility changes only the explanation logs but not the executable action set or violation profile.

F3: Solver-only gains. Performance gains appear only when swapping in a richer controller architecture, and disappear when the same base controller is evaluated under collapsed versus provenance-preserving state semantics.

Any one of these would mean that the tested route failed to isolate process semantics. The response is to revise the active-regime assumptions, visible-context map, dominance order, or certificate preconditions while keeping the solver fixed.

I.2 Ablation logic that isolates novelty

The benchmark in Section H is therefore evaluated under a deliberately strict ablation matrix:

State semantics	Admissibility semantics	Controller family
Collapsed $\mathrm{Agg}(C_t)$	no certification gate	fixed base controller
Collapsed $\mathrm{Agg}(C_t)$	certification gate if definable	same fixed base controller
Provenance-preserving C_t	no certification gate	same fixed base controller
Provenance-preserving C_t	certification gate	same fixed base controller

The non-stacking read is justified only if the shift from the first row to the fourth row cannot be reproduced by architecture changes alone, and if at least part of the gain already appears when moving from collapsed to provenance-preserving state semantics under the same solver.

I.3 Interpretive rule

The controller is treated as an implementation witness, not as the contribution. If the same base solver behaves differently because the environment state now contains contradiction-bearing provenance and the admissibility relation now depends on certification, that supports the process-class claim. If all gains require a different network stack, then the result remains a "better controller" result rather than a process-semantics result.

J Operationalization: dominance-gated EcoFair

The formalism above is an environment class, not a solver prescription. The least risky implementation path is therefore not to invent a wholly new controller stack, but to use the smallest possible operationalization that can witness the new semantics. A contradiction-aware wrapper around EcoFair-CH-MARL is useful precisely because it keeps the solver story conservative while the paper tests the stronger claim that state semantics and admissibility semantics have changed [14].

J.1 Controller decomposition

Let the strategic layer produce budgets, fairness multipliers, or routing envelopes via a policy

$$\pi^S_\Theta(b_t \mid z_t),$$

where z_t denotes strategic observations. Let the operational layer propose physical actions through

$$\pi^O_\theta(a^{\text{base}}_t \mid o_t, b_t),$$

where o_t are local operational observations and b_t are the current strategic boundary conditions. Add a gate policy

$$G_\psi : (x_t, C_t, a^{\text{base}}_t) \mapsto (a^p_t, a^e_t, S_t),$$

where S_t is either a dominant support certificate or the empty set. The gate does not replace EcoFair's strategic and operational logic. It decides whether a proposed critical action can be certified, is replaced by a safe fallback, or must be paired with an epistemic exception act.

J.2 One-step execution template

A practical one-step implementation is:

1. Update the claim base using the latest observations:

$$C_t \leftarrow \mathrm{RBP}\left(C_{t-1} \cup \Gamma(o_t, \varnothing)\right), \qquad \mathcal{K}_t \leftarrow \mathrm{MCC}(C_t).$$

2. Run the strategic planner to obtain current budgets or multipliers b_t.
3. Let the operational controller propose $a^{\text{base}}_t \sim \pi^O_\theta(\cdot \mid o_t, b_t)$.
4. For each critical component of a^{base}_t, search for a certificate in $\mathsf{DCert}_i(x_t, C_t, a^{\text{base}}_i)$ and compute contradiction mass $\kappa_i(x_t, C_t, a^{\text{base}}_i)$.
5. If a dominant support certificate exists and the contradiction mass is below threshold, set

$$a^p_t \leftarrow a^{\text{base}}_t, \qquad a^e_t \leftarrow \mathsf{certify}, \qquad S_t \leftarrow S^\star.$$

6. Otherwise, invoke the gate policy to choose an epistemic exception act, e.g.

$$a_t^e \sim \pi_\psi^E(\cdot \mid x_t, C_t, a_t^{\text{base}}),$$

and pair it with a safe fallback physical action such as hold, defer, or a no-op.

7. Execute (a_t^p, a_t^e) in the simulator, update constraint multipliers using the realized physical transition, and log S_t for audit.

Proposition J.1 (Gate-safety by construction)**.** *Fix a controller of the form described above and suppose the gate G_ψ obeys the following rule on every decision epoch:*

(i) if a proposed critical action $a_i^{\text{base}} \in A_i^{\text{crit}}$ admits some dominant support certificate

$$S \in \mathsf{DCert}_i(x_t, C_t, a_i^{\text{base}})$$

with

$$\kappa_i(x_t, C_t, a_i^{\text{base}}) \leq \eta_i,$$

then the gate may emit that critical action together with $a_i^e \in \mathsf{Exc}_i(a_i^{\text{base}})$ chosen as certify*;*

(ii) otherwise, the gate must replace the proposed critical action by either

(a) a non-critical safe fallback physical action, or

(b) an epistemic exception act paired with a physically safe fallback.

Then every executed joint action is admissible, i.e.

$$\mathcal{F}(x_t, C_t, a_t) = 1$$

for all t.

Proof. If case (i) holds, admissibility follows directly from Definition 6.13: the critical action is supported by a dominant certificate and its contradiction mass is below threshold. If case (ii) holds, the gate does not execute an uncertified critical action. By hypothesis, it instead emits either a non-critical safe fallback or an epistemic exception act paired with such a fallback. Definition 6.13 only imposes certification-or-exception conditions on critical actions. Hence the executed joint action remains admissible. □

Remark J.2 (What gate-safety does not prove)*.* Proposition J.1 is intentionally narrow. It guarantees admissibility by construction, not optimality, minimal intervention, or calibrated escalation behavior. Those stronger properties remain empirical or require additional assumptions.

Proposition J.3 (Frozen-solver bundle transfer)**.** *Fix the strategic and operational layers of the dominance-gated EcoFair wrapper and let π^{gate} denote the resulting policy. Suppose the gate activates on at most M_T decision epochs, where an activation means either blocking an uncertified critical admit or pairing the executed action with an epistemic exception. Assume each activation contributes at most δ_E extra idling-emission units and at most δ_F extra service-disparity units relative to the frozen base action, and contributes at most one unit to intervention spend. Then*

$$V_T(\pi^{\text{gate}}) = 0, \qquad I_T(\pi^{\text{gate}}) \leq M_T,$$

$$E_T(\pi^{\text{gate}}) \le E_T(\pi^{\text{base}}) + \delta_E M_T, \qquad F_T(\pi^{\text{gate}}) \le F_T(\pi^{\text{base}}) + \delta_F M_T.$$

In particular, if $M_T = o(T)$, the gate can enforce zero safety burden while preserving sublinear average intervention, emission, and fairness overhead relative to the frozen base controller.

Proof. Zero safety burden follows from Proposition J.1: the gate never executes an uncertified critical action. By assumption, each activation contributes at most one unit to intervention spend and bounded overhead δ_E, δ_F in the emission and fairness metrics. Summing those bounds over at most M_T activations yields the stated inequalities. □

Proposition J.4 (Frozen-solver transfer to budget overshoot metrics)**.** *Fix budgets $B_E, B_F, B_I \ge 0$ and suppose the dominance-gated wrapper is evaluated with the EcoFair-aligned metrics of Definition F.7. Let π^{base} denote the frozen strategic/operational controller and π^{gate} the wrapped controller. If each gate activation increases cumulative extra idling emissions by at most $\bar{\epsilon}$, increases the chosen disparity statistic by at most $\bar{\phi}$, and contributes at most one unit to intervention spend, then*

$$\mathcal{X}_T^E(\pi^{\text{gate}}) \le \mathcal{X}_T^E(\pi^{\text{base}}) + \bar{\epsilon} M_T,$$

$$\mathcal{X}_T^F(\pi^{\text{gate}}) \le \mathcal{X}_T^F(\pi^{\text{base}}) + \bar{\phi} M_T, \qquad \mathcal{X}_T^I(\pi^{\text{gate}}) \le \mathcal{X}_T^I(\pi^{\text{base}}) + M_T,$$

while still satisfying

$$V_T(\pi^{\text{gate}}) = 0.$$

Thus, under a frozen solver, moving to the provenance-preserving process semantics can enforce zero safety burden with budget overshoot growing at most linearly in the number of gate activations.

Proof. Zero safety burden follows from Proposition J.1. Let $\mathcal{E}_T^{\text{idle}}$ and $\mathcal{D}_T$ denote the actual benchmark-side emission and disparity metrics. By assumption,

$$\mathcal{E}_T^{\text{idle}}(\pi^{\text{gate}}) \le \mathcal{E}_T^{\text{idle}}(\pi^{\text{base}}) + \bar{\epsilon} M_T, \qquad \mathcal{D}_T(\pi^{\text{gate}}) \le \mathcal{D}_T(\pi^{\text{base}}) + \bar{\phi} M_T,$$

and

$$I_T(\pi^{\text{gate}}) \le I_T(\pi^{\text{base}}) + M_T.$$

For any scalar u and budget B, the map $u \mapsto (u - B)_+$ is 1-Lipschitz, so

$$\mathcal{X}_T^E(\pi^{\text{gate}}) = (\mathcal{E}_T^{\text{idle}}(\pi^{\text{gate}}) - B_E)_+ \le (\mathcal{E}_T^{\text{idle}}(\pi^{\text{base}}) - B_E)_+ + \bar{\epsilon} M_T = \mathcal{X}_T^E(\pi^{\text{base}}) + \bar{\epsilon} M_T,$$

and similarly for $\mathcal{X}_T^F$ and $\mathcal{X}_T^I$. □

Remark J.5 (Why the transfer result helps the novelty case)**.** Proposition J.4 still does not make the gate the contribution. Its role is narrower: it shows that once the solver is frozen, any observed safety gain with controlled budget overshoot can be attributed to the changed process semantics rather than to a larger policy class.

J.3 Training options

The safest first experiment is staged training.

1. Train or reuse the EcoFair strategic and operational layers on the original reward and constraint structure.
2. Freeze those layers initially.
3. Train the gate policy G_ψ to minimize contradiction-sensitive regret, violation events, and epistemic intervention cost on top of the frozen base controller.
4. Only after the gate demonstrates separation on Berth-1-Conflict should joint fine-tuning be attempted.

This sequencing keeps the empirical story clean. If the provenance-preserving gate beats aggregated-state baselines on the minimal benchmark, the gain can be attributed to the new process semantics rather than to an uncontrolled architecture change.

J.4 Interpretation of the implementation witness

The dominance-gated EcoFair wrapper is not itself the paper's novelty. It is only an implementation witness for the new process semantics. That distinction matters. If the wrapper fails while the fixed-solver ablations in Section I also fail, then the correct conclusion is that that implementation witness has not isolated process semantics. If the wrapper succeeds but only after major architectural changes, the result is still ambiguous. The cleanest outcome is narrower: a conservative wrapper is enough to expose a difference that already comes from provenance-bearing state and certificate-based feasibility.

K Validation roadmap and refinement rule

At this stage, no additional controller components or result-like claims are introduced. The next work is to validate, narrow, or stress-test the process-semantics claim under a frozen solver, with the experimental lock in Section H.14 treated as binding. The validation sequence is as follows.

1. **Lock the observation model for the separation theorem.** The separation requires the latent physical cases to share the same controller-visible non-claim context $\chi(x)$ while differing in the provenance-bearing claim base. If the benchmark exposes the hidden berth-readiness variable directly, the theorem no longer tests fragmented truth.
2. **Make dominance domain-specific.** Replace the abstract priority map π_L with a maritime priority specification covering local authority, operational context, recency, and optional calibrated source trust. This should be declared before replay experiments, not fitted afterward.
3. **Run the one-berth fixed-solver ablation first.** Compare provenance-preserving and provenance-erased inputs under the same controller, training budget, policy class, reward, and scenario seeds. Do not scale to the 16-port twin until the one-berth separation is clean.

4. **Validate the maritime block key.** The proposed key covers traffic family, weather regime, disruption family, horizon, operating mode, and simulator version. Split any key whose within-block replay diagnostics fail; otherwise the seed-block certificate is too coarse.

5. **Certify the metric lifts.** Ground emissions through the twin-native idling constant. Treat service disparity in the following order: direct service proxy first, witness-free min–max second, APD third, and Gini only when the replay-certified denominator distortion is narrow and stable.

6. **Stress-test dependence and seed shift.** Report blockwise coverage, the finite-key scenario-mixture radius, and a sensitivity curve over the residual within-key shift parameter. The result should show how much coverage is lost under plausible deployment shift.

7. **Keep the solver secondary.** If the gain appears only after changing the controller architecture, learning objective, or policy class, the result is implementation-specific and cannot be used as process-class evidence.

The refinement rule is simple. The process-class framing is justified where the fixed-solver, provenance-preserving condition separates from the provenance-erased condition on feasibility, regret, or bundle burden. Where it does not separate, the result is reported as a boundary case for OS-CMASP: provenance was present but not operationally active under the declared visible context, dominance order, contradiction burden, and replay distribution. This is still a process-semantics result; it is not a reason to recast the work as a controller add-on.

Distilled thesis

The core thesis is that regulated maritime coordination is not always adequately modeled as constrained control over a fused operational state. In disruption-heavy settings, the action-relevant state can include incompatible claims indexed by observer, situation, authority, freshness, and provenance. OS-CMASP makes those claims part of the environment state, makes epistemic repair part of the action space, and makes support certification or escalation part of critical-action feasibility. The central novelty test is solver-independent: erasing observer–situation provenance can force regret, feasibility loss, or budget burden in an active-regime benchmark where an observer-aware policy avoids that burden under the same visible non-claim context and the same solver family. Emission and fairness claims remain secondary unless they are carried by simulator-grounded replay certificates rather than by additional controller machinery.

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